

# A FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

ABSTRACT. A countable discrete group is MF if it embeds in the unitary group of a matrix quotient  $\mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$ ; equivalently, if it admits finite-dimensional approximate unitary representations, asymptotically multiplicative in operator norm, that separate every nontrivial element from the identity. We construct an explicit finitely presented group  $E$  that is not MF, refuting the conjecture that every countable group is MF.

More generally, let  $\Gamma$  be a finitely presented property-(T) group with a proper injective endomorphism  $\alpha$ . From  $(\Gamma, \alpha)$  we build a finitely presented group with a nontrivial central involution  $w$  such that  $\Theta(w) = 1$  for every homomorphism  $\Theta$  to  $\mathcal{U}(\mathcal{Q})$ . In every operator-norm asymptotic representation, one-sided conjugation preserves the asymptotic commutant of  $\Gamma$  in normalized Hilbert–Schmidt norm; if  $\Theta(w) \neq 1$ , compression to the  $(-1)$ -spectral projection of  $\Theta(w)$  contradicts this. Taking  $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  and  $\alpha(v, A) = (2v, A)$  gives  $E$ , on eight generators and forty-one relators.

The group  $E$  is sofic, hence hyperlinear, so neither soficity nor hyperlinearity implies the MF property; its reduced  $C^*$ -algebra  $C_r^*(E)$  is separable, exact and stably finite but not MF; and  $E/\langle w \rangle$  is not MF either, so  $\{1, w\}$  is a proper subgroup of the kernel common to all homomorphisms  $E \rightarrow \mathcal{U}(\mathcal{Q})$ .

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## 1. INTRODUCTION

Can every countable discrete group be realized by matrices, if multiplication is only required to hold up to operator-norm errors that vanish along a sequence of finite dimensions? Shulman calls the positive assertion the *MF conjecture* [Sh], and no group was known to fail it [Sc, Sh]. We prove that the answer is no: there is an explicit finitely presented group, on eight generators and forty-one relators, that is not MF. Residually finite groups satisfy the conjecture through their finite quotients, as do LEF groups [CDE] and, by quasidiagonality [TWW], all amenable groups. Recent work establishes it for many amalgamated free products [Sc, Sh, GKEMP].

**The result.**

**Theorem A** (an explicit finitely presented non-MF group). *Let  $E$  be the group on the eight generators  $v_1, v_2, v_3, x, y, z, t, c$  with the forty-one relators displayed in Definition 6.1, and put*

$$w = [tct^{-1}, v_1(tct^{-1})v_1^{-1}].$$

*The defining relations make  $w$  central, and with  $c^2 = 1$  centrality forces  $w^2 = 1$ .*

- (1)  *$w \neq 1$  in  $E$ , so  $w$  is a central involution;*
- (2) *for every sequence  $(d_n)$  of positive integers and every homomorphism  $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$  into the unitary group of the corresponding matrix quotient, one has  $\Theta(w) = 1$ .*

*In particular  $E$  is not MF, and neither  $C_{\max}^*(E)$  nor  $C_r^*(E)$  is an MF  $C^*$ -algebra. A Tietze reduction presents the same group on six generators with thirty-two relators (Section S1).*

The proof is in Section 6.

**Outline of the proof.** An inclusion between two objects of equal finite size is an equality.

$$(1) \quad \begin{array}{ccc} X \subseteq Y, |X| = |Y| & V \subseteq W, \dim V = \dim W & \begin{array}{c} p \leq q, p \sim q \\ \text{in a finite } C^*\text{-algebra} \end{array} \\ \Downarrow & \Downarrow & \Downarrow \\ X = Y & V = W & p = q. \end{array}$$

Here  $p \sim q$  is Murray–von Neumann equivalence, and a  $C^*$ -algebra is *finite* when every isometry in it is unitary. The three columns are the arguments for finite quotients, for finite-dimensional representations, and for  $\mathcal{Q}$ . Operator norm is the approximation norm throughout, and the normalized Hilbert–Schmidt norm appears only as the norm of the Hilbert space on which the conjugation representation acts.

Let  $\pi$  be a homomorphism to a finite group and let  $t\Gamma t^{-1} \subseteq \Gamma$ . Then  $\pi(t\Gamma t^{-1}) = \pi(\Gamma)$ : the left side is a conjugate of the right side contained in it, and the two are finite of the same cardinality. So an element  $c$  centralizing  $\Gamma$

has  $\pi(tct^{-1})$  centralizing  $\pi(\Gamma)$ , and every commutator  $[tct^{-1}, a]$  with  $a \in \Gamma$  lies in  $\ker \pi$ .

Every finite-dimensional representation  $\pi$  of the ambient group over every field gives the same conclusion, with the commutant in place of the subgroup. Let  $C$  be the commutant of  $\pi(\Gamma)$  in  $\text{End}_k(V)$ . The inclusion  $t\Gamma t^{-1} \subseteq \Gamma$  implies  $C \subseteq \pi(t)C\pi(t)^{-1}$ , the two sides have equal finite dimension, and therefore

$$C = \pi(t)C\pi(t)^{-1}$$

(Theorem 3.3 and Remark S6.2). Since  $\pi(c) \in C$ , also  $\pi(tct^{-1}) \in C$ , so it commutes with every  $\pi(a)$ ,  $a \in \Gamma$ .

Regard  $M_d(\mathbb{C})$  as the Hilbert space  $L^2(M_d(\mathbb{C}), \text{tr}_d)$ , on which conjugation by a unitary acts unitarily. For unitaries  $u_1, u_2$ , uniformly in  $d$ ,

$$\|\text{Ad } u_1 - \text{Ad } u_2\|_{B(L^2(M_d(\mathbb{C}), \text{tr}_d))} \leq 2 \|u_1 - u_2\|,$$

so an operator-norm asymptotic representation induces one by conjugation on the Hilbert–Schmidt spaces.

Let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and let  $t, c \in H$  satisfy

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

In a norm ultraproduct  $B_\omega$  of the blocks, the classes  $\pi(g) = [\text{Ad } U_n(g)]_\omega$  define a unitary representation of  $H$ . By property (T), the spectrum of the Hermitian average  $h$  of the generators is bounded away from 1 off the invariant vectors, so the spectral projection of  $h$  at 1 lies in  $B_\omega$  and projects onto the  $\iota(\Gamma)$ -invariant vectors; call it  $P$ . Writing  $V = \pi(t)$ , the inclusion above implies  $P \leq VPV^*$ ; the two projections are Murray–von Neumann equivalent, and  $B_\omega$  is finite, so  $P = VPV^*$ . Hence conjugation by  $t$  preserves the asymptotic commutant of  $\iota(\Gamma)$  in normalized Hilbert–Schmidt norm (Theorem 3.4).

Put  $d = tct^{-1}$  and  $u = [d, \iota(a)]$  for  $a \in \Gamma$ . Since  $c$  commutes with  $\iota(\Gamma)$ , its lifts lie in that asymptotic commutant, so by Theorem 3.4  $\| [U_n(d), U_n(\iota(\gamma))] \|_2 \rightarrow 0$  for every  $\gamma \in \Gamma$ . Consequently, in every operator-norm asymptotic representation  $(U_n)$  of  $H$ ,

$$\|U_n(u) - 1\|_2 \rightarrow 0.$$

Suppose now that  $w = u^2$  is central in  $H$  with  $w^2 = 1$ , and let  $\Theta$  be a homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  with  $\Theta(w) \neq 1$ . Then  $q = \frac{1}{2}(1 - \Theta(w))$  is a nonzero projection commuting with  $\Theta(H)$ , and rounding the coordinate averages of unitary lifts represents  $q$  by projections  $q_n \in M_{d_n}(\mathbb{C})$ , nonzero for infinitely many  $n$ . Compressing to  $q_n$ , and giving each corner its own normalized trace, yields an operator-norm asymptotic representation of  $H$  in which  $w$  goes to  $-1$ . There the display above gives  $\|U_n(u)^2 - 1\|_2 \rightarrow 0$ , while  $\|-1 - 1\|_2 = 2$ . Hence  $\Theta(w) = 1$ , and if  $w \neq 1$  then  $H$  is not MF (Theorem 4.1).

Let  $\Gamma$  be a finitely presented Kazhdan group with a proper injective endomorphism  $\alpha$ , let  $a \in \Gamma \setminus \alpha(\Gamma)$ , and adjoin  $t$  and  $c$  subject to  $t\gamma t^{-1} = \alpha(\gamma)$ ,

$c^2 = 1$  and  $[c, \gamma] = 1$ . In the direct limit of  $\alpha$  the cosets  $t\Gamma$  and  $at\Gamma$  are distinct, since  $t^{-1}at \in \Gamma$  would give  $a \in \alpha(\Gamma)$ . Let the ambient group permute the generators of a Clifford group indexed by that coset space, and send  $c$  to the generator at the base coset; then  $d$  and  $ada^{-1}$  map to the generators at those two cosets. Distinct Clifford generators anticommute, so  $u$  maps to a product of two anticommuting involutions and  $w = u^2$  to  $-1$  (Theorem 5.1).

Taking

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I)$$

gives the group  $E$  of Theorem A.

Only the subgroup  $\iota(\Gamma)$  is assumed to have property (T). The group  $E$  surjects onto  $\mathbb{Z}$ , by sending  $t$  to 1 and every other generator to 0, so  $E$  itself does not have property (T).

The reduced group  $C^*$ -algebra  $C_r^*(E)$  is separable and stably finite but not MF (Theorem C); it is also exact. The group  $E$  is sofic (Theorem D), so neither soficity nor hyperlinearity implies the MF property. The quotient  $E/\langle w \rangle$  is not MF either, so  $\{1, w\}$  is a proper subgroup of the MF residual of  $E$  (Theorem B). The non-MF argument by itself decides neither soficity nor nonsocicity of  $E$  (Section S13).

**Relation to prior work.** The corresponding  $C^*$ -algebra question is older. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable  $C^*$ -algebra was known not to be MF [BK, CDE]. Their natural candidate was  $C_r^*(F_2)$ , which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative solution of the Connes embedding problem subsequently supplied abstract stably finite non-MF algebras [JNVWY, FGH]. No countable group was known whose reduced or maximal group  $C^*$ -algebra fails to be MF [Sh].

In his 2018 ICM address, Thom asked for which Schatten  $p$ -norms there exist non-approximable groups [Th, Question 3.11]. The case  $p = 2$  was settled by De Chiffre–Glebsky–Lubotzky–Thom [DGLT], the cases  $1 < p < \infty$  by Lubotzky–Oppenheim [LuO], and the case  $p = 1$  by Bachner–Dogon–Lubotzky [BDL]. Three approximation problems were left by that work: whether every countable group is sofic (permutations in the Hamming metric), hyperlinear (unitaries in the normalized Hilbert–Schmidt norm), and MF (unitaries in operator norm). The first was answered negatively in 2026: a nonsoc group was constructed in [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. The present paper answers the third negatively; at the time of writing the hyperlinear problem is open. Our construction begins, as those arguments do, with a relation  $t\Gamma t^{-1} \subseteq \Gamma$  for a property-(T) subgroup  $\Gamma$ . Here the approximation is by unitaries in operator norm, and conjugation by  $t$  preserves the asymptotic commutant of  $\Gamma$  in normalized Hilbert–Schmidt norm in every operator-norm asymptotic representation. Throughout, “MF” is the operator-norm property of Carrión–Dadarlat–Eckhardt [CDE], in

the form used by Shulman and by Bachner–Dogon–Lubotzky [Sh, BDL]; Section 2 fixes it and its equivalent formulations.

The closest analogue is the recent work on nonsofic groups. One-sided conjugation of a property-(T) subgroup, together with an element centralizing that subgroup, appears in [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; Alekseev–Thom prove a related centralizer rigidity for sofic embeddings [AT]. Those results concern Hamming or tracial approximations. Here the group acts by conjugation on the Hilbert–Schmidt space of each matrix block, and finiteness of the norm ultraproduct as a  $C^*$ -algebra forces the projection onto the  $\iota(\Gamma)$ -fixed subspace to equal its unitary conjugate (Theorem 3.4).

Slofstra’s hyperlinear-profile construction uses a Clifford group, a distinguished central involution, a shift, and an HNN doubling map [Slo18, Section 2 and Remark 3.3]. His quotients by the elements sent to the identity by every exact, and by every approximate, finite-dimensional representation [Slo17, Definitions 2.5–2.7] are the exact and the approximate analogues of the MF residual used below. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Fritz’s marked  $BS(2, 3)$  example is the context for the cyclic comparison of Section S7 [Fri, SV]. Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedures used in Section S5 [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5], and handle a finite normal subgroup by a character-isotypic corner [BDL, proof of Proposition 1.6]; Theorem S5.3 combines that corner with one-sided conjugation of a property-(T) subgroup.

Bekka and Bekka–Valette study property (T) through the conjugation representation  $\pi \otimes \bar{\pi}$  on Hilbert–Schmidt space [Be, BV], and Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. The group-MF framework is that of Carrión–Dadarlat–Eckhardt, who also give, through Abels’ group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18].

The group property is weaker than the algebra property. For the algebra one asks for an embedding  $C_r^*(E) \hookrightarrow \mathcal{Q}$  of  $C^*$ -algebras; for the group, only for an injective homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$ , which need not extend to the algebra. Failure of MF for a group  $C^*$ -algebra therefore leaves open whether the group itself embeds. For  $E$ , Theorem A rules out both.

## 2. MATRIX QUOTIENTS

All groups are discrete and countable. We use the standard formulation of property (T) for unitary representations on complex Hilbert spaces. For group elements  $[g, h] = ghg^{-1}h^{-1}$ ; for operators  $[x, y]_- = xy - yx$ . A  $C^*$ -homomorphism is unital only when this is stated; in particular an MF embedding may be nonunital. For a unital  $C^*$ -algebra  $A$  we write  $\mathcal{U}(A)$  for

its unitary group. On  $M_r(\mathbb{C})$  we use the normalized trace  $\text{tr}_r$ , the operator norm  $\|\cdot\|$ , and the normalized Hilbert–Schmidt norm  $\|x\|_2 = \text{tr}_r(x^*x)^{1/2}$ . For a nonzero projection  $q \in M_r(\mathbb{C})$  we equip the corner  $qM_r(\mathbb{C})q$  with its own normalized trace, and where the trace has to be named we write  $\|x\|_{2,\text{tr}_r}$  for the same quantity. The unnormalized trace is  $\text{Tr} = r \cdot \text{tr}_r$  on  $M_r(\mathbb{C})$ . We use without comment:  $\|x\|_2 \leq \|x\|$ ;  $|\text{tr}_r(x)| \leq \|x\|$ ; and  $\|uxv\|_2 = \|x\|_2$  for unitaries  $u, v$ .

Given a sequence  $(d_n)_{n \in \mathbb{N}}$  of positive integers, write

$$(2) \quad \mathcal{Q}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}} M_{d_n}(\mathbb{C}),$$

where the product is the bounded  $C^*$ -product and  $\bigoplus_n$  is the  $c_0$ -sum, the ideal of sequences with  $\|x_n\| \rightarrow 0$ ; equivalently,  $\mathcal{Q}$  is the corona algebra of  $\bigoplus_n M_{d_n}(\mathbb{C})$ . Write  $q: \prod_n M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$  for the quotient map. Following Blackadar–Kirchberg [BK], a separable  $C^*$ -algebra is  $MF$  if it embeds into an algebra of the form (2).

The following lifting lemmas are standard perturbation theory; see Loring [Lo] for a systematic treatment.

**Lemma 2.1** (lifting). *Every unitary  $u \in \mathcal{Q}$  lifts to a sequence of unitaries.*

*Proof.* Lift  $u$  to a bounded sequence  $(x_n)$ ; unitarity of  $q((x_n))$  gives  $\|x_n^*x_n - 1\| \rightarrow 0$ . Once  $\|x_n^*x_n - 1\| \leq \frac{1}{2}$ , the matrix  $x_n$  is invertible and the polar correction  $u_n = x_n(x_n^*x_n)^{-1/2}$  is unitary with  $\|u_n - x_n\| \leq 2\|x_n\|\|x_n^*x_n - 1\| \rightarrow 0$  by continuous functional calculus; set  $u_n = 1$  at the finitely many remaining indices.  $\square$

Write

$$(3) \quad \mathcal{N} = \{(u_n) \in \prod_n \mathcal{U}(d_n) : \|u_n - 1\| \rightarrow 0\},$$

a normal subgroup of  $\prod_n \mathcal{U}(d_n)$ .

**Lemma 2.2** (unitaries in the quotient). *For every sequence  $(d_n)$ , with  $\mathcal{N}$  as in (3), the map*

$$\kappa_{(d_n)}: \prod_n \mathcal{U}(d_n) / \mathcal{N} \longrightarrow \mathcal{U}(\mathcal{Q}), \quad [(u_n)] \longmapsto q((u_n)),$$

*is a canonical group isomorphism. A homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  is therefore equivalent to the following data: a sequence of unitaries for each element of  $H$ , multiplicative up to operator-norm errors tending to 0, taken modulo sequences converging to 1.*

*Proof.* The kernel of  $(u_n) \mapsto q((u_n))$  is exactly  $\mathcal{N}$ , so the induced map is injective. Surjectivity is Lemma 2.1.  $\square$

An *operator-norm asymptotic representation* of  $H$  is a sequence of maps  $U_n: H \rightarrow \mathcal{U}(d_n)$  such that  $\|U_n(g)U_n(h) - U_n(gh)\| \rightarrow 0$  for all  $g, h \in H$ . Taking  $g = h = 1$  gives  $\|U_n(1) - 1\| \rightarrow 0$  automatically, since  $U_n(1)$  is unitary. By Lemma 2.2, choosing unitary lifts  $U_n(g)$  of  $\Theta(g)$  turns a homomorphism

$\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  into such a sequence, and every such sequence arises this way. A countable group is *MF* if it admits an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$  for some sequence of dimensions.

**Proposition 2.3** (Equivalent formulations of the MF property). *For every countable group, the group-MF definition above — an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$  — is equivalent to the definition by sequences of unitaries. Both are equivalent to the existence of approximate unitary representations with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

Section S3 defines the approximate unitary representations and proves the proposition. Two stronger conventions occur in the literature and are not used here: in [GKEMP] the approximating maps must also recover the canonical trace and the left-regular norm on every group-ring element, and the purely matricial field (PMF) property of [MdlS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

### 3. ONE-SIDED CONJUGATION IN MATRIX MODELS

Theorem 3.3 uses the middle implication of (1) for the commutant  $C$  of the base inside a finite-dimensional endomorphism algebra, where  $C \subseteq \pi(t)C\pi(t)^{-1}$  and the two sides have equal finite dimension. The proof of Theorem 3.4 uses the right-hand implication for the projection  $P$  onto the Kazhdan fixed space inside a norm ultraproduct: there  $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$  gives  $P \leq VPV^*$ , the two projections are Murray–von Neumann equivalent, and the ultraproduct is finite, so  $P = VPV^*$ .

**Lemma 3.1** (comparison in a finite algebra). *Let  $A$  be a finite unital  $C^*$ -algebra and let  $p, q \in A$  be projections with  $p \leq q$  and  $p \sim q$ . Then  $p = q$ .*

*Proof.* Let  $r \in A$  satisfy  $r^*r = q$  and  $rr^* = p$ . From  $r^*r = q$  we get  $r = rq$ . Since  $p \leq q$  we have  $qp = p$ , so

$$pr = rr^*r = rq = r, \quad qr = q(pr) = (qp)r = pr = r,$$

and  $r = qr$  as well. Hence the cross terms in  $\sigma^*\sigma$  vanish for  $\sigma = r + (1 - q)$ , and  $\sigma^*\sigma = q + (1 - q) = 1$ , so  $\sigma$  is an isometry. Finiteness makes  $\sigma$  unitary, and  $\sigma\sigma^* = p + (1 - q) = 1$  gives  $p = q$ .  $\square$

**Lemma 3.2** (finiteness of a norm quotient of matrix blocks). *Let  $(K_n)$  be finite-dimensional Hilbert spaces and let*

$$B_c = \prod_n B(K_n) \Big/ \bigoplus_n B(K_n)$$

*be the quotient of the bounded family algebra by the  $c_0$ -sum. Then  $B_c$  is a finite  $C^*$ -algebra: every  $\sigma \in B_c$  with  $\sigma^*\sigma = 1$  satisfies  $\sigma\sigma^* = 1$ .*



*Proof.* Lift  $\sigma$  to a bounded family  $(\sigma_n)$ . Then  $\|\sigma_n^* \sigma_n - 1\| \rightarrow 0$ , so for all large  $n$  the Gram defect is at most  $\frac{1}{2}$  and  $\sigma_n^* \sigma_n$  is invertible; each  $\sigma_n$  acts on a finite-dimensional space, so polar correction replaces it by a unitary  $w_n = \sigma_n (\sigma_n^* \sigma_n)^{-1/2}$  with  $\|\sigma_n - w_n\| \rightarrow 0$ . Hence  $\sigma$  is the class of  $(w_n)$ , and  $w_n w_n^* = 1$  at every such  $n$  gives  $\sigma \sigma^* = 1$ .  $\square$

Let  $\omega$  be a free ultrafilter on  $\mathbb{N}$ , let  $K_\omega$  be the Hilbert-space ultraproduct of the  $K_n$ , and let  $B_\omega = \prod_\omega B(K_n)$  be the norm ultraproduct, acting on  $K_\omega$  by  $[A_n]_\omega [\xi_n]_\omega = [A_n \xi_n]_\omega$ . The proof above applies to  $B_\omega$  with “for all large  $n$ ” replaced by “on a set belonging to  $\omega$ ”, so  $B_\omega$  is finite too. Its action on  $K_\omega$  is faithful: if  $A = [A_n]_\omega \neq 0$  then  $\lim_\omega \|A_n\| = \delta > 0$ , and unit vectors  $\xi_n$  with  $\|A_n \xi_n\| \geq \|A_n\| - 1/n$  give  $\|A[\xi_n]_\omega\| = \delta$ . Hence for projections  $P, Q \in B_\omega$  the inclusion  $\text{ran } P \subseteq \text{ran } Q$  makes  $QP$  and  $P$  act identically on  $K_\omega$ , so  $QP = P$ , that is,  $P \leq Q$ ; the converse is immediate.

### The exact case.

**Theorem 3.3** (the finite-dimensional case). *Let  $k$  be a field,  $V$  a finite-dimensional  $k$ -vector space,  $H$  a group,  $\Gamma \leq H$  a subgroup, and  $t, c \in H$ ,  $a \in \Gamma$  elements with  $t\Gamma t^{-1} \subseteq \Gamma$  and  $c\gamma = \gamma c$  for all  $\gamma \in \Gamma$ . Then every homomorphism  $\pi: H \rightarrow \text{GL}(V)$  satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

More generally, every such  $\pi$  is trivial on the subgroup  $\mathfrak{D}(H, \Gamma)$  of (10).

*Proof.* Let

$$C = \{x \in \text{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of  $\pi(\Gamma)$ , a linear subspace of  $\text{End}_k(V)$ , and write  $\Phi(x) = \pi(t)x\pi(t)^{-1}$  for conjugation by  $\pi(t)$ , a linear automorphism of  $\text{End}_k(V)$ .

We claim  $C \subseteq \Phi(C)$ . Let  $x \in C$  and put  $y = \pi(t)^{-1}x\pi(t)$ , so that  $x = \Phi(y)$ . For  $\gamma \in \Gamma$  we have  $t\gamma t^{-1} \in \Gamma$ , hence  $\pi(t\gamma t^{-1})$  commutes with  $x$ , and therefore

$$\pi(\gamma)y = \pi(t)^{-1}\pi(t\gamma t^{-1})x\pi(t) = \pi(t)^{-1}x\pi(t\gamma t^{-1})\pi(t) = y\pi(\gamma).$$

Thus  $y \in C$ , proving the claim. Since  $\Phi$  is a linear isomorphism,  $\dim_k \Phi(C) = \dim_k C < \infty$ , and the inclusion  $C \subseteq \Phi(C)$  forces

$$(4) \quad C = \Phi(C) = \pi(t)C\pi(t)^{-1}.$$

Now  $\pi(c) \in C$  because  $c$  centralizes  $\Gamma$ , so  $\pi(tct^{-1}) = \Phi(\pi(c)) \in C$  by (4). Since  $a \in \Gamma$ , the operator  $\pi(tct^{-1})$  commutes with  $\pi(a)$ , so  $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$  and

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1.$$

Section S6 proves the subgroup-valued assertion from the same calculation.  $\square$

In  $E$ , every finite-dimensional linear representation over every field, and hence every homomorphism to a finite group, maps the distinguished word to the identity (Corollary S6.1).

An MF model is only asymptotically multiplicative, so its coordinate fixed spaces are not invariant subspaces. Operator-norm control keeps the adjoint actions almost multiplicative on the Hilbert–Schmidt spaces, so the Kazhdan projection lies in the norm ultraproduct, where finiteness replaces the equality of dimensions used above. Property (T) cannot be dropped: the group  $E_{\text{BS}}$  of Section S7 has cyclic base, its distinguished word is trivial in every finite-dimensional representation over every field, and some homomorphism  $E_{\text{BS}} \rightarrow \mathcal{U}(\mathcal{Q})$  maps that word to a nontrivial element.

**Theorem 3.4** (one-sided conjugation preserves asymptotic commutants). *Let  $\Gamma$  and  $H$  be groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, let  $s \in H$ , and let  $(d_n)$  be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

*Let  $U_n: H \rightarrow \mathcal{U}(d_n)$  be an operator-norm asymptotic representation. If  $(x_n)$  is uniformly operator-norm bounded and asymptotically centralizes  $\iota(\Gamma)$  in normalized Hilbert–Schmidt norm,*

$$\|x_n U_n(\iota(\gamma)) - U_n(\iota(\gamma)) x_n\|_2 \rightarrow 0 \quad (\gamma \in \Gamma),$$

*then the conjugated sequence does as well:*

$$\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma))]\|_2 \rightarrow 0 \quad (\gamma \in \Gamma).$$

*Proof.* Suppose the conclusion fails: there are  $\gamma_0 \in \Gamma$ ,  $\delta > 0$ , and an infinite set  $I \subseteq \mathbb{N}$  with  $\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma_0))]\|_2 \geq \delta$  for  $n \in I$ . Fix a free ultrafilter  $\omega$  on  $\mathbb{N}$  with  $I \in \omega$ .

*The adjoint model.* Regard  $M_{d_n}(\mathbb{C})$  as the Hilbert space  $K_n = L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$  and let each group element act on it by conjugation,  $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$ . Operator-norm almost multiplicativity of  $U_n$  makes  $g \mapsto \text{Ad } U_n(g)$  an operator-norm asymptotic representation on these Hilbert spaces.

*The ultraproduct.* Let  $K_\omega$  and  $B_\omega = \prod_\omega B(K_n)$  be as in Section 3; write  $[\cdot]_\omega$  for classes in either. The action of  $B_\omega$  on  $K_\omega$  is faithful,  $B_\omega$  is finite — every isometry in it is unitary — and an inclusion of ranges in  $K_\omega$  is an inequality of projections in  $B_\omega$ . Asymptotic multiplicativity makes

$$\pi: H \rightarrow \mathcal{U}(B_\omega), \quad \pi(g) = [\text{Ad } U_n(g)]_\omega,$$

a homomorphism.

*The projection onto the fixed subspace.* Property (T) provides a finite symmetric generating set  $S \subseteq \Gamma$  with  $1 \in S$  and a Kazhdan constant  $\kappa \in (0, 1]$  for the pair  $(S, \kappa)$ . Let  $h$  be the Hermitian part of  $\frac{1}{|S|} \sum_{s' \in S} \pi(\iota(s'))$  and let  $\text{Fix} \subseteq K_\omega$  be the subspace of vectors fixed by every  $\pi(\iota(\gamma))$ . Both  $\text{Fix}$  and its orthocomplement are invariant under each unitary  $\pi(\iota(s'))$ , hence under  $h$ . On  $\text{Fix}$ ,  $h = 1$ . The restriction of  $\pi \circ \iota$  to  $\text{Fix}^\perp$  has no nonzero invariant

vector, so a unit vector  $\xi \in \text{Fix}^\perp$  satisfies  $\|\pi(\iota(s'))\xi - \xi\| \geq \kappa$  for some generator. For that generator  $\text{Re}\langle \pi(\iota(s'))\xi, \xi \rangle \leq 1 - \kappa^2/2$ , while every other term is at most 1. Averaging gives  $\langle h\xi, \xi \rangle \leq 1 - \kappa^2/(2|S|)$ . Consequently

$$\text{sp}(h) \subseteq \left[-1, 1 - \frac{\kappa^2}{2|S|}\right] \cup \{1\},$$

and the spectral projection  $P$  of  $h$  at the isolated point 1 belongs to  $B_\omega$  by continuous functional calculus and projects onto  $\text{Fix}$ .

*The two projections agree.* Let  $V = \pi(s)$  and  $Q = V P V^*$ . Since  $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$ , every vector of  $\text{Fix}$  is fixed by each  $\pi(s\iota(\gamma)s^{-1})$ ; and  $\eta$  is fixed by all  $\pi(s\iota(\gamma)s^{-1})$  exactly when  $V^*\eta \in \text{Fix}$ . Hence  $\text{ran } P = \text{Fix} \subseteq V \text{Fix} = \text{ran } Q$ , so  $P \leq Q$ . The element  $r = V^*Q$  satisfies  $r^*r = Q$  and  $rr^* = V^*QV = P$ , so  $P \sim Q$ . Since  $B_\omega$  is finite, Lemma 3.1 gives  $Q = P$ .

*Conclusion.* Let  $\xi = [\xi_n]_\omega \in K_\omega$ , where  $\xi_n \in K_n$  is the vector of  $x_n$ ; the uniform operator-norm bound makes  $\xi$  well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each  $\pi(\iota(\gamma))$  fixes  $\xi$ . Thus  $\xi \in \text{Fix}$ . Since  $Q = P$ , we obtain  $V\xi \in V \text{Fix} = \text{ran } Q = \text{ran } P = \text{Fix}$ . The vector of  $U_n(s)x_n U_n(s)^*$  is  $\text{Ad } U_n(s) \xi_n$ , so, again by unitary invariance,

$$\| [U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma))] - \|_2 = \| (\text{Ad } U_n(\iota(\gamma)) - 1) \text{Ad } U_n(s) \xi_n \| \longrightarrow 0$$

along  $\omega$ , for every  $\gamma \in \Gamma$ . For  $\gamma = \gamma_0$  this contradicts  $I \in \omega$ .  $\square$

#### 4. THE CENTRAL INVOLUTION OBSTRUCTION

Let  $w$  be a central involution of  $H$  and let  $\Theta$  be a homomorphism to  $\mathcal{U}(\mathcal{Q})$  with  $\Theta(w) \neq 1$ . Then  $\Theta(w)$  is a self-adjoint unitary commuting with  $\Theta(H)$ , the projection  $\frac{1}{2}(1 - \Theta(w))$  is nonzero, and compressing to it gives a matrix model of  $H$  in which  $w$  equals  $-1$  and each block has its own normalized trace.

On the corner  $q_n M_{d_n}(\mathbb{C}) q_n$  the operator norm is the restriction of the ambient one, so the compressed and polar-corrected maps remain operator-norm asymptotically multiplicative, while the trace is the corner's own normalized trace  $\text{tr}_{r_n}$ , not the restriction of  $\text{tr}_{d_n}$ . Hence  $\|1 - (-1)\|_{2, \text{tr}_{r_n}} = 2$  for every value of  $r_n/d_n$ . Compression converts a normalized Hilbert–Schmidt bound into an operator-norm obstruction: padding an asymptotic representation with an identity block leaves every operator-norm quantity unchanged while sending every normalized Hilbert–Schmidt distance to zero (Section S13).

**Theorem 4.1** (central involution obstruction). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

*Put  $d = tct^{-1}$  and  $u = [d, \iota(a)]$  for some  $a \in \Gamma$ . If  $w := u^2$  is a nontrivial central involution of  $H$ , then every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  maps  $w$  to the identity, and  $H$  is not MF.*

*Proof.* Let  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  be a homomorphism and suppose  $\Theta(w) \neq 1$ .

*The  $(-1)$ -spectral corner.* Choose coordinate unitary lifts  $(V_{g,n})$  of  $\Theta$  (Lemma 2.2) and put  $p_n = \frac{1}{2}(1 + V_{w,n})$ , the averaging operator of the two-element subgroup  $F = \{1, w\}$ . Because the lifts are asymptotically multiplicative and  $w^2 = 1$ , the Hermitian part of  $p_n$  is asymptotically idempotent, so its spectrum clusters at  $\{0, 1\}$  and rounding at  $\frac{1}{2}$  by continuous functional calculus produces genuine projections at operator-norm distance  $o(1)$  from  $p_n$ ; centrality of  $w$  makes them asymptotically central for all of  $\Theta(H)$ . Let  $q_n$  be their complements. If  $q_n$  vanished for all large  $n$  the averages would converge to 1 and  $\Theta(w)$  would be 1; so  $q_n \neq 0$  along infinitely many coordinates, which we retain and relabel by  $\mathbb{N}$ . Compressing every lift to  $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$  and polar-correcting the compressions — each is almost unitary in the corner, by asymptotic centrality — gives an operator-norm asymptotic representation  $(W_{g,n})$  of  $H$  on nonzero blocks with

$$\|W_{w,n} + 1\| \rightarrow 0.$$

Equip each corner block with its own normalized trace  $\text{tr}_{r_n}$ ; all estimates below are relative to  $r_n$  and independent of the ratio  $r_n/d_n$ .

*Transport on the corner.* Since  $c$  centralizes  $\iota(\Gamma)$ , the sequence  $(W_{c,n})$  centralizes  $(W_{\iota(\gamma),n})$  asymptotically in operator norm, hence in normalized Hilbert–Schmidt norm. By Theorem 3.4 applied to  $x_n = W_{c,n}$ , the conjugated sequence  $(W_{d,n})$  lies in the same asymptotic commutant. Therefore  $\|W_{u,n} - 1\|_2 \rightarrow 0$ , and squaring gives  $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$ .

*The contradiction.* The relation  $w = u^2$  gives  $\|W_{u,n}^2 - W_{w,n}\| \rightarrow 0$ , while  $W_{w,n} \rightarrow -1$  in operator norm and therefore also in normalized Hilbert–Schmidt norm. Hence  $\|W_{u,n}^2 + 1\|_2 \rightarrow 0$  and  $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$ , so  $\|2 \cdot 1\|_2 \rightarrow 0$ , which is false. Thus  $\Theta(w) = 1$ . Since  $w \neq 1$ , every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  has nontrivial kernel, so  $H$  is not MF.  $\square$

The same conclusion holds for an arbitrary finite normal subgroup of  $N_{\text{conj}}$  (Theorem S5.3), and for an arbitrary normal property-(T) subgroup  $K \leq N_{\text{conj}}$ , with a Kazhdan spectral projection in place of the two-element average; there  $K$  may be noncentral and torsion-free (Theorem S5.6).

## 5. THE KAZHDAN–CLIFFORD CONSTRUCTION

**Theorem 5.1** (Kazhdan–Clifford construction). *Let  $\Gamma$  be a finitely presented group with property (T), let  $\alpha: \Gamma \hookrightarrow \Gamma$  be an injective endomorphism, and choose  $a \in \Gamma \setminus \alpha(\Gamma)$ . Adjoin elements  $t, c$  and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \quad c^2 = 1, \quad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

*Put*

$$d = tct^{-1}, \quad u = [d, a], \quad w = u^2 = [d, ada^{-1}],$$

and impose centrality of  $w$ . More explicitly, for a finite generating set  $S_\Gamma$  it is enough to impose the conjugation and commutation relations for  $s \in S_\Gamma$ , together with

$$[w, s] = 1 \quad (s \in S_\Gamma), \quad [w, t] = [w, c] = 1.$$

Thus  $w$  commutes with a generating set of the entire resulting group. The group  $E(\Gamma, \alpha, a)$  is finitely presented, the canonical map  $\Gamma \rightarrow E(\Gamma, \alpha, a)$  is injective, and  $w$  is a nontrivial central involution. Every homomorphism  $E(\Gamma, \alpha, a) \rightarrow \mathcal{U}(\mathcal{Q})$  maps  $w$  to 1, so  $E(\Gamma, \alpha, a)$  is not MF.

*Proof.* Finite presentability follows from a relative finite presentation: start with the free product of  $\Gamma$  and the free group on  $t, c$ , and quotient by the finitely many displayed relations for a finite generating set of  $\Gamma$ .

The relation  $w^2 = 1$  is redundant. Put  $b = ada^{-1}$ . Both  $d$  and  $b$  are involutions,  $w = [d, b] = (db)^2$ , and direct cancellation gives

$$(5) \quad dwd^{-1} = w^{-1}.$$

Centrality also gives  $dwd^{-1} = w$ , whence  $w = w^{-1}$  and  $w^2 = 1$ .

For nontriviality, realize the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

concretely. Let  $\Gamma_\infty$  be the direct limit of the system  $\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \dots$ , whose  $n$ th copy of  $\Gamma$  we call its  $n$ th level; injectivity of  $\alpha$  passes to the limit, so the level-zero map embeds  $\Gamma$  in  $\Gamma_\infty$ . Applying  $\alpha$  levelwise induces an injective endomorphism  $\sigma$  of  $\Gamma_\infty$ . It is surjective, hence an automorphism, because an element at level  $k$  equals  $\sigma$  of the same element at level  $k+1$ . Put  $G = \Gamma_\infty \rtimes_\sigma \mathbb{Z}$ , with stable letter  $t$  implementing  $\sigma$ ; then  $t\gamma t^{-1} = \alpha(\gamma)$  for  $\gamma \in \Gamma$ . Write  $X = G/\Gamma$  for the coset space of the level-zero copy. Let  $G$  permute the Clifford generators  $(e_x)_{x \in X}$  and map  $c$  to  $e_\Gamma$ . Then

$$d \mapsto e_{t\Gamma}, \quad ada^{-1} \mapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give  $t^{-1}at \in \Gamma$ , equivalently  $a \in \alpha(\Gamma)$ . Hence the two Clifford generators anticommute and

$$w \mapsto (e_{t\Gamma}e_{at\Gamma})^2 = -1.$$

The Clifford sign is central, so the map respects every defining relation. It follows both that  $w \neq 1$  and that the canonical map of  $\Gamma$  is injective.

Thus  $w = u^2$  is a nontrivial central involution, and by Theorem 4.1 every homomorphism to  $\mathcal{U}(\mathcal{Q})$  maps  $w$  to the identity. Since  $w \neq 1$ , every such homomorphism has nontrivial kernel.  $\square$

*Remark 5.2* (a small-cancellation realization). Fournier-Facio has pointed out (personal communication, 2026) that the affine base of Sections S1–S2 can be replaced by a small-cancellation construction as in [FFF]. His finitely presented torsion-free property-(T) group  $P$  contains a copy of every finitely presented torsion-free group, in particular a direct product  $P_1 \times P_2$  with  $P_i \cong P$ . Then  $\alpha: P \xrightarrow{\cong} P_1 \leq P$ , together with any  $a \in P_2 \setminus \{1\}$ , satisfies the

hypotheses of Theorem 5.1. The group  $E(P, \alpha, a)$  has torsion even though  $P$  does not, since the construction adjoins the involution  $c$ .

## 6. A FINITELY PRESENTED NON-MF GROUP

Specialize Theorem 5.1 to

$$(6) \quad \Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I).$$

The affine group  $\Gamma$  is finitely presented and has property (T) [BHV, Example 1.7.4(i)], and Section S1 identifies it with the presentation used below. The endomorphism  $\alpha$  is injective because doubling is injective on  $\mathbb{Z}^3$ , it is proper because  $e_1 \notin 2\mathbb{Z}^3$ , and  $a \notin \alpha(\Gamma)$  for the same reason. Theorem 5.1 therefore applies, and relative to  $\Gamma$  the resulting group has the finite presentation

$$E = \left\langle \Gamma, t, c \mid \begin{array}{l} t\gamma t^{-1} = \alpha(\gamma), \quad [c, \gamma] = 1 \quad (\gamma \in S_\Gamma), \\ c^2 = 1, \quad [w, g] = 1 \end{array} \right\rangle,$$

where  $S_\Gamma$  is a finite generating set of  $\Gamma$ ,  $g$  runs over the generators, and  $w = [tct^{-1}, a(tct^{-1})a^{-1}]$ . Expanding a standard presentation of the affine group in the generators  $v_1, v_2, v_3$  for the translations and  $x, y, z$  for the linear part turns this into the eight-generator, forty-one-relator presentation below. First define the *linear presentation*

$$(7) \quad \mathcal{R} = \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

and the *base*

$$(8) \quad \mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \mid \begin{array}{l} \text{the eight relations in (7),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

Write  $\iota: \mathcal{B} \rightarrow E$  for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

**Definition 6.1.** Use the six base generators and the relations displayed in (8). Define

$$E = \left\langle v_1, v_2, v_3, x, y, z, t, c \left| \begin{array}{l} \text{(B)} \quad \text{all relations in (8),} \\ \text{(T)} \quad tv_i t^{-1} = v_i^2 \ (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ \text{(C)} \quad c^2 = 1, \\ [c, g] = 1 \ (g \in \{v_1, v_2, v_3, x, y, z\}), \\ \text{(W)} \quad [w, g] = 1 \ (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right. \right\rangle,$$

where  $w$  abbreviates the word  $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$ . This is an explicit finite presentation with eight generators and forty-one relators: twenty in family (B), six in family (T), seven in family (C), and eight in family (W). Family (B) is a presentation of the base; family (T) conjugates each base generator by  $t$  into the base; family (C) makes  $c$  an involution commuting with the base generators; and family (W) makes  $w$  central. Centrality together with  $c^2 = 1$  implies  $w^2 = 1$  by (5). Families (T) and (C) give, respectively,

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \quad [c, \iota(\mathcal{B})] = 1.$$

A Tietze reduction removes two generators and nine relators (Section S1).

**Lemma 6.2** (the distinguished word is a square). *Let  $H_1$  be a group and  $d, a_1 \in H_1$  with  $d^2 = 1$ . Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

*Proof.* Put  $b = a_1 d a_1^{-1}$ , so  $b^2 = a_1 d^2 a_1^{-1} = 1$ ; thus  $b^{-1} = b$  and  $d^{-1} = d$ . Then  $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d (a_1 d a_1^{-1}) = db$ , and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

*Proof of Theorem A.* That  $w \neq 1$  is Proposition S2.3.

Let  $(d_n)$  be given, write  $\mathcal{Q} = \mathcal{Q}((d_n))$ , and let  $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$  be a homomorphism. Apply the central involution obstruction (Theorem 4.1) with  $\Gamma = \mathcal{B}$ , the canonical homomorphism  $\iota$ , the elements  $t, c \in E$ , and  $a = v_1$ . Its hypotheses are the displayed relators: Proposition S1.1 gives property (T) of  $\mathcal{B}$ ; family (T) gives  $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ ; and family (C) gives  $[c, \iota(\mathcal{B})] = 1$ . In  $E$  we have  $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$ , so Lemma 6.2 with  $a_1 = \iota(v_1)$  gives

$$w = [d, \iota(v_1) d \iota(v_1)^{-1}] = [d, \iota(v_1)]^2 = u^2, \quad u = [d, \iota(v_1)].$$

Centrality of  $w$  is a defining relator; with  $c^2 = 1$  it gives  $w^2 = 1$  by (5); and  $w \neq 1$  as above. Hence  $\Theta(w) = 1$ .

An MF embedding of  $E$  would be an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$ , and every homomorphism to  $\mathcal{U}(\mathcal{Q})$  maps the nontrivial element  $w$  to 1; so  $E$

is not MF. Suppose  $C_r^*(E)$  were an MF algebra, with an injective, possibly nonunital  $*$ -homomorphism  $e: C_r^*(E) \rightarrow \mathcal{Q}$ . Complement correction gives the injective unitary homomorphism  $u \mapsto e(u) + (1 - e(1))$ ; composing it with the canonical map  $E \rightarrow \mathcal{U}(C_r^*(E))$ , which is injective because  $g \mapsto \lambda_g$  is, exhibits an injective homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$ . The canonical map  $E \rightarrow \mathcal{U}(C_{\max}^*(E))$  is injective as well, since the left regular representation factors through  $C_{\max}^*(E)$  and sends  $u_g$  to  $\lambda_g$ , so an MF embedding of  $C_{\max}^*(E)$  produces an injective homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$  after the same conversion.  $\square$

**Corollary 6.3** (uniform finite operator-norm obstruction). *There exist  $\delta > 0$  and a finite set  $F_0 \subset E$  such that, for every  $d \geq 1$ , every map  $\phi: E \rightarrow \mathcal{U}(d)$  satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \quad (g, h \in F_0)$$

*satisfies  $\|\phi(w) - 1\| < 1$ .*

*Proof.* Fix an exhaustion  $F_1 \subseteq F_2 \subseteq \dots$  of  $E$  by finite sets. If no such pair  $(\delta, F_0)$  exists, then for each  $n$  there is a unitary-valued  $\phi_n$  with multiplicative defect at most  $1/n$  on  $F_n$  and  $\|\phi_n(w) - 1\| \geq 1$ . The sequence  $(\phi_n)$  is an operator-norm asymptotic representation of  $E$ , so by Lemma 2.2 its class is a homomorphism  $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$ , and  $\|\Theta(w) - 1\| \geq 1$  gives  $\Theta(w) \neq 1$ , against Theorem A.  $\square$

The proof is nonquantitative: it gives no bound for  $\delta$  or for  $|F_0|$ . An effective modulus would need an effective Kazhdan constant for the base together with effective versions of the compression estimates.

## 7. CONSEQUENCES OF THE EXAMPLE

Throughout,  $d = tct^{-1}$  and  $u = [d, \iota(v_1)]$ , so that  $w = u^2$ . Write  $\text{Res}_{\text{MF}}(G)$  for the *MF residual* of a countable group  $G$ : the intersection of the kernels of all its homomorphisms to  $\mathcal{U}(\mathcal{Q})$  (Definition S9.1). Thus  $G$  is MF exactly when  $\text{Res}_{\text{MF}}(G) = 1$  (Proposition S9.3), and Theorem A says  $w \in \text{Res}_{\text{MF}}(E)$  with  $w \neq 1$ .

### The quotient by the central involution.

**Theorem B** (the quotient of  $E$  by  $\langle w \rangle$ ). *The quotient  $E/\langle w \rangle$  is not MF. The image of  $[\iota(v_1), d]$ , the inverse of the distinguished commutator  $u$ , is nontrivial there and lies in  $\text{Res}_{\text{MF}}(E/\langle w \rangle)$ ; consequently  $\{1, w\}$  is a proper subgroup of  $\text{Res}_{\text{MF}}(E)$ .*

The proof is in Section S10. If an involution is centralized by the conjugate copy of a Kazhdan subgroup, and its conjugates under that subgroup commute with one another, then every homomorphism to  $\mathcal{U}(\mathcal{Q})$  identifies the entire conjugation orbit (Theorem S10.3). In  $E$  the pairwise commutators of those conjugates of  $d$  lie in  $\langle w \rangle$ , so in  $E/\langle w \rangle$  the element  $d$  satisfies these hypotheses, and every homomorphism  $E/\langle w \rangle \rightarrow \mathcal{U}(\mathcal{Q})$  maps  $u$  to the identity. Thus  $E/\langle w \rangle$  is a second finitely presented non-MF group, and Theorem A does not determine  $\text{Res}_{\text{MF}}(E)$  (Section S13).



**A stably finite non-MF group algebra.** Every MF algebra is stably finite [BK]; the converse fails. The canonical trace on  $C_r^*(E)$  is faithful, so  $C_r^*(E)$  is stably finite (Lemma S11.1), independently of the MF question. Abstract examples follow from the failure of the Connes embedding problem [FGH]; the reduced algebra of the finitely presented group  $E$  is an explicit one (Theorem C). For  $C_r^*(\mathrm{SL}_3(\mathbb{Z}))$  and  $C_r^*(\mathrm{SL}_4(\mathbb{Z}))$  the MF property is undecided; only the stronger PMF property is known to fail, for  $\mathrm{SL}_4(\mathbb{Z})$  [MdlS].

**Theorem C** (the reduced group algebra of  $E$ ).  *$C_r^*(E)$  is a unital separable  $C^*$ -algebra which has a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

*Proof.*  $C_r^*(E)$  is unital and separable because  $E$  is countable, and by Lemma S11.1 the canonical trace is faithful and the algebra is therefore stably finite. It is not an MF algebra by Theorem A.  $\square$

The algebra is also exact, by the block normal form of  $E$  and permanence for free products and extensions (Section S11). The algebra is not nuclear: the canonical map  $\mathcal{B} \rightarrow E$  is injective (Section S2), so  $E$  contains an infinite Kazhdan group and is nonamenable, whence  $C_r^*(E)$  is not nuclear [Lan]. The nuclear form of the Blackadar–Kirchberg problem — whether every stably finite separable nuclear  $C^*$ -algebra is quasidiagonal — therefore remains open.

### A sofic example.

**Theorem D** (a finitely presented sofic non-MF group). *The group  $E$  of Definition 6.1 is sofic, hence hyperlinear. Consequently  $E$  is itself a finitely presented sofic group that is not MF, and a finitely presented hyperlinear group that is not MF. It is also sofic and not LEF, and in particular not residually finite.*

The proof is in Section S11.3, through the decomposition  $E \cong C(\mathcal{G}) \rtimes \Sigma$  of Proposition S11.2: here  $\mathcal{G}$  is the graph on  $X$  whose cliques are the blocks of the coset space, and  $W$  is the quotient of  $E$  in which lamps at sites of different blocks anticommute.

That quotient is the Clifford group  $W = \mathrm{Cl}(X) \rtimes G$  over the coset space  $X = G/\Gamma$  of the ascending HNN extension  $G = \Gamma_\infty \rtimes \mathbb{Z}$  used in the proof of Theorem 5.1, at the affine data (6); Section S2 realizes it over  $\mathbb{Q}$ , writing  $\Sigma$  for  $G$ . The surjection  $E \rightarrow W$  sends  $w$  to the nontrivial Clifford sign  $\zeta$ , so functoriality (Lemma S9.2) gives  $\zeta \in \mathrm{Res}_{\mathrm{MF}}(W)$  and  $W$  is not MF. The extension  $1 \rightarrow \mathrm{Cl}(X) \rightarrow W \rightarrow \Sigma \rightarrow 1$  has locally finite kernel and residually finite quotient, hence MF kernel and MF quotient, while  $W$  is not MF: the MF property is not closed under extensions (Section S11). Corollary S9.8 gives the corresponding failure for quotients, through  $F_8 \twoheadrightarrow E$ .

## Supplementary material

The supplement has four parts. Sections S1–S3 verify the explicit group and fix the formulations the main text uses. Sections S4–S7 develop the transport argument in the generality it has, and show that property (T) cannot be dropped. Sections S8–S12 draw the consequences. Sections S13 and S14 describe what the method does not decide, and record the maximal group  $C^*$ -algebra.

### S1. THE BASE, AND A SMALLER PRESENTATION

The base  $\mathcal{B}$  of (8) is the affine group  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ , for which property (T) is classical [BHV, Example 1.7.4(i)].

Six generators suffice. The group  $E$  is generated by the six elements  $v_1, x, y, z, t, c$ , and has a Tietze-equivalent presentation on them with thirty-two relators. In (9), eliminate  $v_3 = xv_1x^{-1}$  and  $v_2 = x^2v_1x^{-2}$  by Tietze substitutions and delete the two defining relations; the remaining  $x$ -action relation follows from  $x^3 = 1$ , and the stable-letter,  $c$ -commutation, and centrality relations for  $v_2, v_3$  follow from those for  $v_1$  together with  $[t, x] = 1$  and  $x^3 = 1$ . Deleting these nine leaves thirty-two relators on  $v_1, x, y, z, t, c$ .

Write  $d = tct^{-1}$ . The displayed relations give  $[d, t\iota(\mathcal{B})t^{-1}] = 1$  and leave  $[d, \iota(v_1)d\iota(v_1)^{-1}]$  unconstrained (Figure 1).

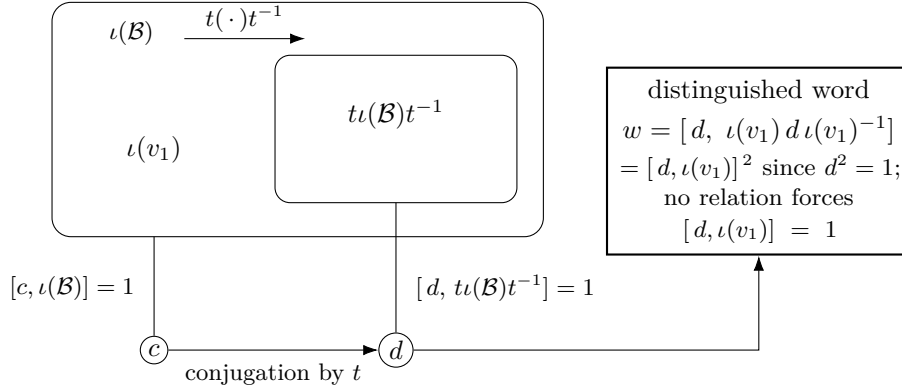


Figure 1: The one-sided conjugation configuration. The relations of (9) give  $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$  and  $[c, \iota(\mathcal{B})] = 1$ , hence  $[d, t\iota(\mathcal{B})t^{-1}] = 1$  for  $d = tct^{-1}$ ; they do not give  $[d, \iota(v_1)] = 1$ , and  $w = [d, \iota(v_1)d\iota(v_1)^{-1}] = [\iota(v_1)]^2$ . Section S2 proves that the inclusion is strict and that  $\iota(v_1)$  lies outside the conjugate copy, as drawn; Theorem 4.1 uses only the inclusion.

#### S1.1. Property (T) of the base.

**Proposition S1.1.** *The twenty-relator group  $\mathcal{B}$  has property (T), in both the real-orthogonal and complex-unitary formulations.*

*Proof.* By Remark S1.2 the presentation (8) defines  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ , which has property (T) [BHV, Example 1.7.4(i)]. For an orthogonal representation  $\rho$  on a real Hilbert space  $H_{\mathbb{R}}$ , the complexification  $H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$  carries the unitary representation  $\rho \otimes 1$ , whose invariant vectors and displacement norms are those of  $\rho$  in each of the two real coordinates; a Kazhdan pair for the unitary formulation is therefore one for the orthogonal formulation.  $\square$

*Remark S1.2* (the classical identification). The eight relations (7) present  $\mathrm{SL}_3(\mathbb{Z})$  [CRV, Theorem 2], with the matrix-labelled generators also recorded in [CLV], and the base  $\mathcal{B}$  is then the affine group  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ : the surjection  $\mathcal{B} \rightarrow \bar{\Gamma}$  of Lemma S2.2 onto the matrix realization is an isomorphism.

## S2. NONTRIVIALITY OF $w$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra's hyperlinear-profile construction [Slo18, Section 2].

**Construction S2.1** (Clifford groups). For a set  $X$ , use the real Clifford-algebra convention generated by  $(c_x)_{x \in X}$  with  $c_x^2 = 1$  and  $c_x c_y = -c_y c_x$  for  $x \neq y$ , and let  $\mathrm{Cl}(X)$  be the subgroup of units generated by  $-1$  and the  $c_x$ . Write  $\zeta = -1$ . In this concrete group,

$$\zeta^2 = c_x^2 = 1, \quad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign  $\zeta$  is nontrivial. Thus  $\mathrm{Cl}(X)$  is a central extension, by the sign  $\zeta$ , of the elementary abelian 2-group on the points of  $X$ : the commutator of two distinct generators is  $\zeta$ , where the generators of a classical lamplighter commute. Every permutation of  $X$  acts on this group by  $c_x \mapsto c_{\sigma(x)}$  and fixes  $\zeta$ .

**Lemma S2.2** (linear model). *For  $M \in \mathrm{GL}_3(\mathbb{Q})$  and  $u \in \mathbb{Q}^3$ , write*

$$A(M, u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

*Assign  $v_i \mapsto A(I_3, e_i)$  and*

$$\begin{aligned} x &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), & y &\mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right), \\ z &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right). \end{aligned}$$

*Let  $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$  be the subgroup generated by these six matrices, and put  $D = \mathrm{diag}(2, 2, 2, 1)$ . The six matrices satisfy all twenty relators of  $\mathcal{B}$ , and*

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of  $\bar{\Gamma}$  which squares the three translation generators and fixes  $x, y, z$ . Moreover the matrix assigned to  $v_1$  does not lie in  $\text{range}(\bar{\alpha})$ .

*Proof.* The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by  $D$  gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in  $\text{GL}_4(\mathbb{Z})$ . Hence  $\bar{\Gamma} \leq \text{GL}_4(\mathbb{Z})$ . However  $D^{-1}\bar{v}_1D$  has entry  $1/2$  in position  $(1, 4)$ , so it is not in  $\bar{\Gamma}$ . This proves the last assertion.  $\square$

Form the direct limit  $T = \varinjlim(\bar{\Gamma}, \bar{\alpha})$ , the direct limit of  $\bar{\alpha}$ , extend it by its shift  $\tau$ , and call the resulting group  $\Sigma = T \rtimes \mathbb{Z}$ . Let  $j: \bar{\Gamma} \hookrightarrow \Sigma$  be the level-zero map. Put  $X = \Sigma/j(\bar{\Gamma})$ , with base point  $o = j(\bar{\Gamma})$ . Since  $D^{-1}\bar{v}_1D \notin \bar{\Gamma}$  (Lemma S2.2), we have  $\tau o \neq j(\bar{v}_1)\tau o$ , and the Clifford generators at these two points anticommute.

**Proposition S2.3.**  $w \neq 1$  in  $E$ .

*Proof.* Form  $W := \text{Cl}(X) \rtimes \Sigma$  as in Construction S2.1. Define a map on the generators of (9) by

$$\begin{aligned} g &\mapsto j(\bar{g}) \in \Sigma \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto \tau, \quad c \mapsto c_o. \end{aligned}$$

The exact checks in Lemma S2.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes  $o$ , so  $c_o$  commutes with every base generator, and  $c_o^2 = 1$ . Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \quad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two points are distinct by Lemma S2.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since  $\zeta$  is central in  $W$ , the remaining relations  $[w, g] = 1$  for  $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$  are also satisfied. Hence the assignment extends to a group homomorphism  $E \rightarrow W$  sending  $w$  to  $\zeta \neq 1$ , and therefore  $w \neq 1$  in  $E$ .  $\square$

When  $X$  is infinite so is  $\text{Cl}(X)$ , but each of its elements involves only finitely many Clifford generators, and every element of  $\Sigma$  used above is a rational  $4 \times 4$  matrix. The two points are distinct because  $D^{-1}\bar{v}_1D$  has the entry  $1/2$  in position  $(1, 4)$  and therefore does not lie in  $\bar{\Gamma} \leq \text{GL}_4(\mathbb{Z})$ .

### S3. THE MF EQUIVALENCES

An *approximate unitary representation* for a countable group  $G$ , a finite set  $F \subseteq G$ , a separation constant  $\delta > 0$ , and a tolerance  $\varepsilon > 0$  is a map  $V: G \rightarrow \mathcal{U}(r)$ , for some positive dimension  $r$ , such that

$$\|V(g)V(h) - V(gh)\| \leq \varepsilon \quad (g, h \in F),$$

$$\|V(g) - V(h)\| \geq \delta \quad (g \neq h \in F).$$

The local formulation in Proposition 2.3 asks for such a model with  $\delta = 1$  for every finite  $F$  and every  $\varepsilon > 0$ .

*Proof of Proposition 2.3.* Lemma 2.2 identifies  $\prod_n \mathcal{U}(d_n)/\mathcal{N}$  with  $\mathcal{U}(\mathcal{Q})$ , which is the first equivalence once the dimension convention is normalized.

*Dimension normalization.* Given positive dimensions  $d_n$ , replace the  $n$ th sequence of unitary lifts by

$$\tilde{V}_n(g) = I_{\mathbb{C}^{d_0}} \oplus \cdots \oplus I_{\mathbb{C}^{d_{n-1}}} \oplus V_n(g)$$

on  $\mathbb{C}^{d_0} \oplus \cdots \oplus \mathbb{C}^{d_n}$ , of strictly increasing dimension  $D_n = d_0 + \cdots + d_n$ . Block-diagonal operator norms are maxima and every earlier block is the same identity for all group elements, so for all  $g, h$

$$\begin{aligned} \|\tilde{V}_n(g)\tilde{V}_n(h) - \tilde{V}_n(gh)\| &= \|V_n(g)V_n(h) - V_n(gh)\|, \\ \|\tilde{V}_n(g) - \tilde{V}_n(h)\| &= \|V_n(g) - V_n(h)\|. \end{aligned}$$

The defect and separation data are therefore unchanged, so the classes  $[(\tilde{V}_n(g))_n]$  define a homomorphism to  $\mathcal{U}(\mathcal{Q}((D_n)))$  — a different matrix quotient, so not the same class — which is injective for exactly the elements the original one separated. The converse is immediate.

Suppose first that  $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$  is injective, and choose unitary lifts  $u_n(g)$  of  $\rho(g)$ . Fix a finite set  $F \subseteq G$  and  $\varepsilon > 0$ . For each ordered pair  $g \neq h$  in  $F$ , injectivity says that  $(u_n(g)^{-1}u_n(h))_n$  does not converge to the identity in operator norm. Hence there are  $\delta_{g,h} > 0$  and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

Choose an integer  $N_{g,h}$  with  $8 < N_{g,h} \delta_{g,h}^2$ . The following estimate then applies: if  $u, v \in \mathcal{U}(d)$  satisfy  $\|u - v\| \geq \delta$ , then  $\|u^{\otimes p} - v^{\otimes p}\| \geq 1$  for some  $p \leq N$  whenever  $8 < N\delta^2$ . Indeed, put  $z = u^*v$ , so that  $\|u - v\| = \|1 - z\|$  and  $z$  has a spectral value  $e^{2\pi i\alpha}$  with  $|1 - e^{2\pi i\alpha}| = 2\sin(\pi\alpha') \geq \delta$ , where  $\alpha' = \min(\{\alpha\}, 1 - \{\alpha\})$ ; since  $(u^{\otimes p})^*v^{\otimes p} = z^{\otimes p}$  has  $e^{2\pi ip\alpha}$  in its spectrum,

$$\|u^{\otimes p} - v^{\otimes p}\| \geq |1 - e^{2\pi ip\alpha}| = 2\sin(\pi \min(\{p\alpha\}, 1 - \{p\alpha\})),$$

which is at least 1 as soon as  $\{p\alpha\}$  lies in the closed interval  $[1/6, 5/6]$ ; the interval is taken closed because the stepping estimate below can land exactly on an endpoint, where the norm is exactly 1, and separation at least 1 is all that is required. The bound  $2\sin(\pi\alpha') \geq \delta$  gives  $\alpha' \geq \delta/(2\pi)$ ; consecutive multiples of  $\alpha$  move  $\{p\alpha\}$  by  $\alpha'$  in a fixed direction modulo 1, which is less than the length  $2/3$  of that interval, so some  $p \leq \lceil 1/(6\alpha') \rceil \leq \pi/(3\delta) + 1 < N$  lands in it. The quotient homomorphism law makes all multiplicative defects for pairs in  $F$  arbitrarily small at every sufficiently large coordinate. Choose, for each  $g \neq h$ , a coordinate with the displayed separation and with every required defect at most  $\varepsilon/N_{g,h}$ . Then the  $p$ th tensor power separates the designated pair by at least 1, while the tensor-product telescoping identity

bounds each new defect by  $p\varepsilon/N_{g,h} \leq \varepsilon$ . Finally take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in  $F$  is separated by at least 1 and every required product has defect at most  $\varepsilon$ . This gives the normalized approximate representation. If  $|F| \leq 1$  there are no distinct pairs and hence no blocks, and the dimension is required to be positive; in that case take  $r = 1$  and the map sending every element of  $G$  to  $1 \in \mathcal{U}(1)$ , which is exactly multiplicative and separates the empty set of pairs.

Conversely, suppose approximate unitary representations with separation constant 1 exist for all finite sets and tolerances. Choose models along an exhaustion with errors tending to zero. Each fixed pair  $g, h$  is then eventually represented multiplicatively, and each fixed pair  $g \neq h$  is eventually separated by 1. Hence the classes define an injective homomorphism into  $\prod_n \mathcal{U}(r_n)/\mathcal{N}$ , with no condition on the dimension sequence. In particular every nonidentity element is separated by some homomorphism to a matrix quotient, so the MF residual of Definition S9.1 is trivial, and the block-sum diagonalization in the proof of Proposition S9.3(2) then assembles the separating sequences into a single faithful homomorphism to  $\mathcal{U}(\mathcal{Q})$ ; that diagonalization uses only the dimension normalization above.

□

## S4. TRANSPORT AT ARBITRARY WEIGHTS

The transport theorem extends to arbitrary weights  $\nu_n$  in place of the dimension; Theorem 3.4 is the case  $\nu_n = d_n$ .

**Theorem S4.1** (transport at an arbitrary weight, and of intertwiners). *Let  $\Gamma$ ,  $H$ ,  $\iota$ , and  $s$  be as in Theorem 3.4.*

- (1) *Let  $(U_n)$  be as there, let  $(\nu_n)$  be nonnegative weights, and suppose there is  $C$  with  $\text{Tr}(x_n^* x_n) \leq C \nu_n$  for all  $n$ , and that for every  $\gamma \in \Gamma$  and every  $\varepsilon > 0$ , eventually*

$$\text{Tr} |x_n - U_n(\iota(\gamma)) x_n U_n(\iota(\gamma))^*|^2 \leq \varepsilon \nu_n.$$

*Then  $U_n(s) x_n U_n(s)^*$  and  $U_n(s)^* x_n U_n(s)$  satisfy both conditions as well.*

- (2) *Let  $U_n^1: H \rightarrow \mathcal{U}(d_n^1)$  and  $U_n^2: H \rightarrow \mathcal{U}(d_n^2)$  be operator-norm asymptotic representations and let  $(x_n)$  be uniformly operator-norm bounded  $d_n^1 \times d_n^2$  matrices with*

$$\frac{\text{Tr} |x_n - U_n^1(\iota(\gamma)) x_n U_n^2(\iota(\gamma))^*|^2}{d_n^1 + d_n^2} \longrightarrow 0 \quad (\gamma \in \Gamma).$$

*Then  $U_n^1(s) x_n U_n^2(s)^*$  satisfies the same convergence for every  $\gamma \in \Gamma$ .*

*Proof.* (1) Give each coordinate space with  $\nu_n > 0$  the inner product  $\langle x, y \rangle = \text{Tr}(y^* x) / \nu_n$ ; an index with  $\nu_n = 0$  forces  $x_n = 0$ . Conjugation by a unitary is unitary for every weight, and both hypotheses and both conclusions concern only the renormalized norms, so the ultraproduct proof of Theorem 3.4 applies with  $\text{tr}_{d_n}$  replaced by  $\text{Tr}(\cdot) / \nu_n$  and nothing else changed: the bound  $\text{Tr}(x_n^* x_n) \leq C \nu_n$  defines the ultraproduct vector, the adjoint actions remain operator-norm almost multiplicative because the weight does not enter the operator norm, and finiteness of the norm ultraproduct turns  $P \leq V P V^*$  into  $P = V P V^*$ , which gives both directions.

(2) The block sum  $U_n^\oplus = U_n^1 \oplus U_n^2$  is an operator-norm asymptotic representation, since block-diagonal operator norms are maxima. The off-diagonal embedding of  $x_n$  has operator norm  $\|x_n\|$ , its commutator with  $U_n^\oplus(\iota(\gamma))$  is the off-diagonal embedding of the intertwining defect — whose normalized Hilbert–Schmidt norm on the block space has the denominator  $d_n^1 + d_n^2$  — and conjugation by  $U_n^\oplus(s)$  acts on the embedded corner by  $x_n \mapsto U_n^1(s) x_n U_n^2(s)^*$ . Theorem 3.4 applied to  $U_n^\oplus$  transports the embedded sequence. Replacing the denominator  $d_n^1 + d_n^2$  by an arbitrary weight  $\nu_n$  and applying (1) to the block sum instead of Theorem 3.4 gives the intertwiner transport at every weight.  $\square$

Taking  $\nu_n = d_n$  gives Theorem 3.4 with the uniform operator-norm bound relaxed to a uniform bound on  $\text{Tr}(x_n^* x_n) / d_n$ . Taking  $\nu_n = k_n$ , the rank of the projection lift in the proof of Theorem S10.1, gives the normalization used there.

Invariant tensors of type  $(p, q)$  may be identified with Hilbert–Schmidt intertwiners from the  $q$ th to the  $p$ th tensor power. Tensor powers of unitaries are exactly multiplicative and satisfy  $\|A^{\otimes p} - B^{\otimes p}\| \leq p \|A - B\|$ , so tensor powers of an operator-norm asymptotic representation are again operator-norm asymptotic representations, and Theorem S4.1(2) applies to every pair of them: at each fixed tensor type and each weight, a sequence asymptotically invariant for the conjugate copy of the Kazhdan image is already asymptotically invariant for the whole of it.

### S5. OBSTRUCTIONS FROM A ONE-SIDED CONJUGATION DATUM

In each result below a subgroup of  $H$  lies in the kernel of every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$ , because it is at once 2-norm trivial and 2-norm far from the identity on a corner. The hypothesis producing the corner varies:

- Theorem S5.3: a *finite* normal subgroup of  $N_{\text{conj}}$ , the corner being the complement of its rounded averaging operator — the case of the central involution of Section 4;
- Theorem S5.6: a normal property-(T) subgroup of  $N_{\text{conj}}$ , which may be infinite, noncentral and torsion-free, the corner being a spectral projection of a Kazhdan average;
- Theorem S5.7: the same statement with the conjugation datum discarded and its only consequence, 2-norm triviality, taken as the hypothesis;
- Corollary S5.8: that statement applied to the subgroup  $\mathfrak{D}(H, L)$  of (10), which depends on  $L$  alone and not on a choice of  $t$  and  $c$ .

**Definition S5.1** (Kazhdan conjugation datum). Let  $H$  be a countable group. A *Kazhdan conjugation datum* in  $H$  is a tuple  $(\Lambda, \iota, t, c)$ , where  $\Lambda$  is a countable group with property (T),  $\iota: \Lambda \rightarrow H$  is a group homomorphism, not assumed injective, and  $t, c \in H$ , subject to

- (M1)  $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$  for every  $\lambda \in \Lambda$ ;
- (M2)  $c \iota(\lambda) = \iota(\lambda) c$  for every  $\lambda \in \Lambda$ .

Write  $d := tct^{-1}$ . The datum determines the normal closure

$$N_{\text{conj}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

For a finitely generated  $\Lambda$ , conditions (M1)–(M2) are imposed by finitely many relations once generating sets are fixed. The relations make  $c$  commute with  $\iota(\Lambda)$  and  $d$  commute with the subgroup  $t\iota(\Lambda)t^{-1}$ , but they do not constrain the commutators  $[d, \iota(\lambda)]$ ;  $N_{\text{conj}}$  is their normal closure.

**Definition S5.2** (2-norm triviality). Let  $U = (U_n)$  be an operator-norm asymptotic representation of a countable group  $H$  and let  $\omega$  be an ultrafilter refining the cofinite filter. Write

$$K_2^\omega(U) = \{g \in H : \lim_\omega \|U_n(g) - 1\|_2 = 0\},$$



the kernel of the homomorphism induced by  $U$  into the tracial ultraproduct of the matrix blocks along  $\omega$ , and hence a normal subgroup of  $H$ . An element  $g$  is *2-norm trivial* if  $g \in K_2^\omega(U)$  for every such  $U$  and every such  $\omega$ . Convergence of the full sequence,  $\|U_n(g) - 1\|_2 \rightarrow 0$ , gives membership along every  $\omega$  at once.

**Theorem S5.3** (finite-normal obstruction criterion). *Let  $H$  be a countable group with a Kazhdan conjugation datum  $(\Lambda, \iota, t, c)$ , and let  $F \trianglelefteq H$  be a finite normal subgroup with  $F \subseteq N_{\text{conj}}$ . Then every homomorphism  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(f) = 1$  for all  $f \in F$ .*

*Proof.* *2-norm triviality of  $N_{\text{conj}}$ .* Let  $U_n: H \rightarrow \mathcal{U}(d_n)$  be an operator-norm asymptotic representation of  $H$  — in particular, coordinate unitary lifts of a homomorphism to  $\mathcal{U}(\mathcal{Q})$  — write  $U_{g,n} = U_n(g)$ , and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \rightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make  $K_2 \trianglelefteq H$ . Since  $c$  centralizes  $\iota(\Lambda)$ , its lifts belong to the asymptotic commutant in normalized Hilbert–Schmidt norm of  $\iota(\Lambda)$ . By Theorem 3.4 applied to  $x_n = U_{c,n}$ , the lifts of  $d = tct^{-1}$  lie there as well. Therefore  $[d, \iota(\lambda)] \in K_2$  for every  $\lambda \in \Lambda$ ; since  $K_2$  is normal, it contains the normal closure of these commutators:  $N_{\text{conj}} \subseteq K_2$ , and so  $F \subseteq K_2$ . Convergence of the full sequence gives triviality along every ultrafilter refining the cofinite filter, so every element of  $N_{\text{conj}}$  is 2-norm trivial (Definition S5.2).

*The coordinate averaging corner.* Suppose a homomorphism  $\Theta$  to  $\mathcal{U}(\mathcal{Q})$  is nontrivial on  $F$ ; choose coordinate unitary lifts  $(V_{g,n})$  of  $\Theta$  and form the coordinate averaging operators

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and  $F$  is finite,  $(p_n)$  is asymptotically self-adjoint and idempotent, and normality of  $F$  makes it asymptotically central for all of  $\Theta(H)$ . The Hermitian part of  $p_n$  is asymptotically idempotent, so its spectrum clusters at  $\{0, 1\}$  and rounding at  $1/2$  by continuous functional calculus yields genuine projections at operator-norm distance  $o(1)$  from  $p_n$ ; the rounded projections are asymptotically central as well. Let  $q_n$  be their complements. Nontriviality of  $\Theta$  on  $F$  forces  $q_n \neq 0$  along infinitely many coordinates — otherwise the averages converge to 1 and  $\Theta|_F$  is trivial. Retain exactly those coordinates and relabel them by  $\mathbb{N}$ . Compress every lift to the corner  $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$  with  $r_n = \text{rank } q_n > 0$ , and polar-correct the compressed matrices: asymptotic centrality makes each compression  $q_n V_{g,n} q_n$  almost unitary in the corner, so its polar unitary is at operator-norm distance  $o(1)$  and almost-multiplicativity is preserved by the correction. The result is an operator-norm asymptotic representation  $(W_{g,n})$  of  $H$  on the nonzero corner blocks. Since the corner is

the complement of the rounded average, the finite average vanishes there:

$$(*) \quad \left\| \sum_{f \in F} W_{f,n} \right\| \longrightarrow 0 \quad \text{in operator norm.}$$

*The contradiction.* Since  $F \subseteq N_{\text{conj}}$  and  $(W_{g,n})$  is itself an operator-norm asymptotic representation of  $H$ , the first step applies to it: every  $f \in F$  satisfies  $\|W_{f,n} - 1\|_2 \rightarrow 0$ . Since  $F$  is finite, the maximum of these  $|F|$  null sequences is itself null, so

$$\sum_{f \in F} W_{f,n} \longrightarrow |F|1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block with its own normalized trace. This contradicts  $(*)$ , because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence no homomorphism to  $\mathcal{U}(\mathcal{Q})$  is nontrivial on  $F$ .  $\square$

The same conclusion holds with  $N_{\text{conj}}$  replaced by any normal subgroup all of whose elements lie in  $K_2^\omega(U)$  for every operator-norm asymptotic representation  $U$  of  $H$  and every  $\omega$ ; Theorem S5.5 is that statement for  $\mathfrak{D}(H, L)$ .

The central involution obstruction (Theorem 4.1) is the case  $F = \{1, w\}$ , where  $p_n = \frac{1}{2}(1 + V_{w,n})$  and the complementary corner gives a model in which  $w = -1$  exactly.

**S5.1. An intrinsic form.** For a subgroup  $L \leq H$ , let  $G^+(L)$  be generated by the elements  $s$  satisfying  $sLs^{-1} \subseteq L$  — the *one-sided conjugators* of  $L$  — and put

$$(10) \quad \mathfrak{D}(H, L) = \left\langle \left\langle [gzg^{-1}, \ell] : g \in G^+(L), z \in C_H(L), \ell \in L \right\rangle \right\rangle_H,$$

which depends only on the pair  $(H, L)$ , not on a choice of datum.

**Corollary S5.4** (transport along all one-sided conjugators). *Let  $(r_n)$  and  $V_{g,n} \in \mathcal{U}(r_n)$  be an operator-norm asymptotic representation of  $H$ , let  $\Lambda$  have property (T), let  $\iota: \Lambda \rightarrow H$  be a homomorphism, and put  $L = \iota(\Lambda)$ . Suppose that  $(x_n \in M_{r_n}(\mathbb{C}))$  is uniformly bounded in operator norm and belongs to the asymptotic commutant in normalized Hilbert–Schmidt norm of  $L$ :*

$$\left\| x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^* \right\|_2 \longrightarrow 0 \quad (\lambda \in \Lambda).$$

*Then every  $g \in G^+(L)$  acts in both directions on that asymptotic commutant. Explicitly, both*

$$V_{g,n} x_n V_{g,n}^* \quad \text{and} \quad V_{g,n}^* x_n V_{g,n}$$

*satisfy the preceding convergence for every  $\lambda \in \Lambda$ .*

*Proof.* For a one-sided conjugator  $s$ , Theorem 3.4 gives forward transport, and the equality  $P = VPV^*$  established in its proof gives  $V^*PV = P$ , hence  $V^*\text{Fix} = \text{Fix}$  and reverse transport. The elements acting in both directions form a subgroup of  $H$ , which therefore contains  $G^+(L)$ .  $\square$

**Theorem S5.5** (the subgroup  $\mathfrak{D}$  and the MF residual). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and put  $L = \iota(\Gamma)$ . If  $F \trianglelefteq H$  is finite and  $F \leq \mathfrak{D}(H, L)$ , then every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  maps every element of  $F$  to the identity.*

*Proof.* Let  $(U_{h,n})$  be coordinate lifts of an operator-norm asymptotic representation of  $H$  and put  $K_2 = \{h : \|U_{h,n} - 1\|_2 \rightarrow 0\}$ , a normal subgroup of  $H$ . If  $z \in C_H(L)$ , then  $(U_{z,n})$  lies in the asymptotic commutant of  $L$ , and by Corollary S5.4 so does  $(U_{gzg^{-1},n})$  for every  $g \in G^+(L)$ . Hence each generator  $[gzg^{-1}, \ell]$ ,  $\ell \in L$ , displayed in (10) belongs to  $K_2$ ; since  $K_2 \trianglelefteq H$ , it contains their normal closure:  $\mathfrak{D}(H, L) \leq K_2$ . Every element of  $\mathfrak{D}(H, L)$  is therefore 2-norm trivial in every operator-norm asymptotic representation of  $H$ . The proof of Theorem S5.3 uses only that triviality, so it applies to any finite normal  $F \leq \mathfrak{D}(H, L)$ .  $\square$

**S5.2. The normal-Kazhdan form of the obstruction.** A finite group has property (T), and its averaging operator is the projection onto its fixed space. The spectral projection of a Kazhdan average replaces that operator for an arbitrary normal property-(T) subgroup, which may be infinite, noncentral and torsion-free.

**Theorem S5.6** (normal-Kazhdan obstruction). *Let  $H$  be a countable group with a Kazhdan conjugation datum  $(\Lambda, \iota, t, c)$ , and let  $K \trianglelefteq H$  be a normal subgroup with property (T) such that  $K \subseteq N_{\text{conj}}$ . Then every homomorphism  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(k) = 1$  for all  $k \in K$ .*

Finite groups have property (T), so  $K = F$  recovers Theorem S5.3. In Theorem S5.6,  $K$  may be infinite, noncentral, and torsion-free.

*Proof.* The containment  $N_{\text{conj}} \subseteq K_2$  from the proof of Theorem S5.3 holds for any normal subgroup, so every element of  $K$  is 2-norm trivial in every operator-norm asymptotic representation of  $H$ .

Suppose a homomorphism  $\Theta$  to  $\mathcal{U}(\mathcal{Q})$  is nontrivial on  $K$ . Its image  $\bar{H} = \Theta(H)$  is countable, hence MF, and Proposition 2.3, applied along an exhaustion of  $\bar{H}$ , gives an operator-norm asymptotic representation  $(V_{g,n})$  in which every fixed nontrivial element eventually stays at operator-norm distance at least 1 from the identity. These models bound only operator-norm distances; at growing dimension they give no lower bound on normalized Hilbert–Schmidt distances. The datum and the subgroup pass to their  $\Theta$ -images with normality and property (T) preserved, and  $\bar{K} = \Theta(K)$  is nontrivial and lies in the subgroup  $N_{\text{conj}}$  determined by the image datum; it suffices to derive a contradiction for  $\bar{H}$ .

*The Kazhdan corner.* Fix a free ultrafilter  $\omega$ , let  $H_\omega$  be the Hilbert-space ultraproduct of the coordinate spaces  $\mathbb{C}^{d_n}$ , and let  $B_\omega = \prod_\omega B(\mathbb{C}^{d_n})$  act on it as in Section 3, so that the classes  $\pi(g) = [V_{g,n}]_\omega$  define a unitary representation of  $\bar{H}$  on  $H_\omega$ . Fix a finite symmetric generating Kazhdan set

$S \subseteq \bar{K}$  with constant  $\varepsilon_0$ , put  $\theta = 1 - \varepsilon_0^2/(4|S|)$ , and let

$$h = \frac{1}{|S|} \sum_{a \in S} \pi(a) \in B_\omega,$$

self-adjoint because  $S$  is symmetric. A unit vector  $\xi$  has  $\langle h\xi, \xi \rangle = 1$  exactly when  $\pi(a)\xi = \xi$  for every  $a \in S$ , while the Kazhdan pair bounds  $\langle h\xi, \xi \rangle$  by  $\theta$  on the orthogonal complement of  $\text{Fix } \pi(\bar{K})$ . The spectrum of  $h$  therefore lies in  $[-1, \theta] \cup \{1\}$ , the spectral projection  $P = \chi_{\{1\}}(h)$  belongs to  $B_\omega$ , and its range is  $\text{Fix } \pi(\bar{K})$ . Put  $q = 1 - P$ .

If  $q = 0$  then  $\pi(k) = 1$  for every  $k \in \bar{K}$ , against the operator-norm separation of a fixed nontrivial element of  $\bar{K}$ . For  $g \in \bar{H}$  the projection  $\pi(g)P\pi(g)^*$  has range  $\text{Fix } \pi(g\bar{K}g^{-1}) = \text{Fix } \pi(\bar{K})$  by normality, so  $P$  and  $q$  commute with  $\pi(\bar{H})$  exactly. Lift  $q$  to projections  $q_n \in M_{d_n}(\mathbb{C})$ , nonzero along  $\omega$  and with  $\|[q_n, V_{g,n}]_-\| \rightarrow_\omega 0$  for every  $g \in \bar{H}$ .

Convergence along  $\omega$  is not convergence as  $n \rightarrow \infty$ , so pass to a subsequence before compressing. Enumerate  $\bar{H} = \{g_1, g_2, \dots\}$ . For each  $j$  the set

$$A_j = \{n : q_n \neq 0 \text{ and } \|[q_n, V_{g_i,n}]_-\| < 1/j \text{ for all } i \leq j\}$$

is a finite intersection of sets in  $\omega$ , hence lies in  $\omega$  and in particular is infinite. Choose  $n_1 < n_2 < \dots$  with  $n_j \in A_j$  and relabel along  $(n_j)_j$ . On the relabelled sequence  $q_n \neq 0$  for every  $n$  and  $\|[q_n, V_{g,n}]_-\| \rightarrow 0$  in the ordinary sense for every fixed  $g \in \bar{H}$ , since  $g = g_i$  lies in the constraint of  $A_j$  for all  $j \geq i$ . Compressing every model to  $q_n M_{d_n}(\mathbb{C}) q_n$  and polar-correcting now yields an operator-norm asymptotic representation  $(W_{g,n})$  of  $\bar{H}$  on nonzero blocks.

*A 2-norm separation on the corner.* Equip each corner block with its renormalized trace. The corner compression satisfies  $ghq \leq \theta q$ , so at  $\omega$ -most coordinates the average over  $S$  of the real parts of the renormalized traces of the compressed models is at most  $\theta + o(1)$ . Some generator in  $S$  therefore has compressed model with real renormalized trace at most  $\theta + o(1)$ , and so normalized Hilbert–Schmidt distance at least  $\sqrt{2(1-\theta)} - o(1)$  from the corner identity; passing to a subsequence fixes a single such generator  $s_0$ .

*The contradiction.* The corner models form an operator-norm asymptotic representation of  $\bar{H}$ , and  $\bar{K}$  lies in the subgroup  $N_{\text{conj}}$  determined by the image datum, so 2-norm triviality applies to them: every element of  $\bar{K}$  — in particular  $s_0$  — satisfies  $\|W_{s_0,n} - 1\|_2 \rightarrow 0$  on the corner, contradicting the separation just obtained.  $\square$

**Theorem S5.7** (obstruction from 2-norm triviality). *Let  $H$  be a countable group and let  $D \leq H$  be a subgroup every element of which is 2-norm trivial (Definition S5.2). Let  $K \trianglelefteq H$  be a normal subgroup with property (T) such that  $K \leq D$ . Then every homomorphism  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(k) = 1$  for all  $k \in K$ .*

*Proof.* Suppose a homomorphism  $\Theta$  to  $\mathcal{U}(\mathcal{Q})$  is nontrivial on  $K$ . The image  $\bar{H} = \Theta(H)$  is countable and MF, and Proposition 2.3 provides an

operator-norm asymptotic representation of  $\bar{H}$  separating every fixed nontrivial element. Every operator-norm asymptotic representation of  $\bar{H}$  composes with  $\Theta$  to one of  $H$ , so every element of  $\Theta(D)$  is again 2-norm trivial, while  $\bar{K} = \Theta(K) \leq \Theta(D)$  is normal, Kazhdan and nontrivial. The spectral projection  $q_n$ , the 2-norm separation at some  $s_0 \in \bar{K}$ , and the compression constructed in the proof of Theorem S5.6 use only that  $\bar{K}$  is normal, Kazhdan and nontrivial, and 2-norm triviality of  $s_0$  on the corner contradicts that separation.  $\square$

Applied to the intrinsic defect, Theorem S5.7 gives the companion of Theorem S5.5 for a normal property-(T) subgroup:

**Corollary S5.8** (normal-Kazhdan form of the intrinsic residual). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and put  $L = \iota(\Gamma)$ . If  $K \trianglelefteq H$  is a normal subgroup with property (T) and  $K \leq \mathfrak{D}(H, L)$ , then every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  maps every element of  $K$  to the identity.*

*Proof.* By the proof of Theorem S5.5, every element of  $\mathfrak{D}(H, L)$  is 2-norm trivial in every operator-norm asymptotic representation of  $H$ . Theorem S5.7 with  $D = \mathfrak{D}(H, L)$  concludes.  $\square$

The subgroup generated by *all* normal property-(T) subgroups of  $\mathfrak{D}(H, L)$  therefore also lies in the kernel of every homomorphism to  $\mathcal{U}(\mathcal{Q})$ .

## S6. THE FINITE-DIMENSIONAL OBSTRUCTION

Every finite-dimensional representation of  $E$  over every field maps  $w$  to the identity (Theorem 3.3 with  $H = E$ ). So finite-dimensional linear representations do not separate the points of  $E$ , the image of  $w$  is trivial in every finite quotient, and the Clifford quotient of Section S2 is infinite. The central involution  $w$  lies in the kernel of every homomorphism to a finite group, so  $E$  is a group of *Deligne type* in the terminology of [BDL, Definition 1.2]. Such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and  $E$  is such a counterexample for the operator norm.

The same calculation proves the final, subgroup-valued assertion of Theorem 3.3. The elements of  $H$  whose conjugation action preserves  $C$  in both directions form a subgroup. Every one-sided conjugator lies in it by the argument above, hence so does all of  $G^+(\Gamma)$ . Thus, for  $g \in G^+(\Gamma)$  and  $z \in C_H(\Gamma)$ , the operator  $\pi(gzg^{-1})$  still belongs to  $C$  and commutes with every  $\pi(\ell)$ ,  $\ell \in \Gamma$ . Hence every displayed generator in (10) lies in  $\ker \pi$ ; since  $\ker \pi \trianglelefteq H$ , the whole normal closure  $\mathfrak{D}(H, \Gamma)$  does as well.

**Corollary S6.1.** *The group  $E$  of Definition 6.1 admits no injective homomorphism into  $\mathrm{GL}(V)$  for any finite-dimensional vector space  $V$  over any field. More strongly, every finite-dimensional linear representation maps  $w$  to the identity, and every homomorphism to a finite group maps  $w$  to the identity.*

*Proof.* Apply Theorem 3.3 with  $H = E$ , the image of  $\mathcal{B}$ , and the elements  $t, c, v_1$ ; the hypotheses hold by the defining relations (9). Every finite-dimensional representation maps  $w$  to the identity, and  $w \neq 1$  by Proposition S2.3. For a finite target, compose with its faithful left regular representation on the group algebra over  $\mathbb{Q}$ . The composite is a finite-dimensional linear representation, so it maps  $w$  to the identity. Evaluating the resulting identity matrix at the basis vector of the identity element returns the image of  $w$  in the finite group.  $\square$

*Remark S6.2* (a conjugation-invariant size forbids strict compression). The proof of Theorem 3.3 used one feature of finite dimensionality: if  $V \subseteq W$  and  $\dim V = \dim W < \infty$ , then  $V = W$ . Abstractly, let a conjugation-invariant size  $\sigma$  with values in  $\mathbb{N}$  be defined on the subgroups of a group  $Q$ , and suppose  $K \leq L$  with  $\sigma(L) \leq \sigma(K)$  implies  $K = L$ . If  $gKg^{-1} \subseteq K$  then the two have equal size, so the inclusion is an equality: in such a target no subgroup is properly conjugated into itself. Cardinality on the subgroups of a finite group qualifies, which is the finite-quotient argument of Section 1.

The finite-dimensional case is *not* that principle applied to subgroups. The natural candidate size, the dimension  $\dim_k \text{End}_H(V)$  of the commutant, is order-reversing rather than order-preserving, and it does not separate nested subgroups:  $A_4 < S_4$  in the standard three-dimensional representation have the same commutant  $k \cdot 1$ . Theorem 3.3 applies the principle to the commutant instead, where the inclusion  $C \subseteq uCu^{-1}$  between subspaces of equal finite dimension is an equality (4); it is that equality, and not an equality of subgroup images, that sends the defect to the identity.

## S7. THE CYCLIC COMPARISON

**Theorem S7.1** (the cyclic comparison). *Define the presentation*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where  $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$ . Then  $E_{\text{BS}}$  is finitely presented and  $w_{\text{BS}} \neq 1$ . There is an explicit surjective homomorphism from  $E_{\text{BS}}$  onto the subgroup of the Clifford semidirect product generated by the displayed images, and it maps  $w_{\text{BS}}$  to the nontrivial central sign. Nevertheless every finite-dimensional representation of  $E_{\text{BS}}$  over every field maps  $w_{\text{BS}}$  to the identity.

So a distinguished word can be trivial in every exact finite-dimensional representation without being trivial in the group.

*Proof of Theorem S7.1.* As with  $w$ , the relator  $w_{\text{BS}}^2 = 1$  is redundant:  $tct^{-1}$  and  $\gamma_0(tct^{-1})\gamma_0^{-1}$  are involutions, so centrality forces it by (5). Deleting it gives a six-relator presentation of the same group. The stable relation gives  $t\langle \gamma_0 \rangle t^{-1} \subseteq \langle \gamma_0 \rangle$ , and  $c$  centralizes this cyclic base. By the commutant argument of Theorem 3.3, every finite-dimensional representation over every field maps  $w_{\text{BS}}$  to the identity.

For nontriviality, use the affine realization of  $BS(1, 2)$  and its coset action. The base point is fixed by the cyclic base, while the two points represented by  $t$  and  $\gamma_0 t$  are distinct. Attach Clifford generators to these points and map  $c$  to the generator at the base point. All seven relators hold, and

$$w_{BS} \mapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus  $w_{BS} \neq 1$ . Restricting the target to the subgroup generated by the three displayed images gives the asserted surjection onto a Clifford quotient.  $\square$

Sharpness of the Kazhdan hypothesis. The affine group  $BS(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$  is solvable and the Clifford group on its coset space is locally finite, so their semidirect product is amenable. The Clifford quotient in Theorem S7.1 is a subgroup of it, hence amenable and, by [TWW], MF; its distinguished word is the nontrivial Clifford sign by the calculation above. Composing the quotient map with a faithful homomorphism to  $\mathcal{U}(\mathcal{Q})$  gives  $\Theta: E_{BS} \rightarrow \mathcal{U}(\mathcal{Q})$  with  $\Theta(w_{BS}) \neq 1$ , so property (T) cannot be omitted from the central involution obstruction; in the residual language of Section S9,  $w_{BS} \notin \text{Res}_{MF}(E_{BS})$ .

The homomorphism to  $\mathcal{U}(\mathcal{Q})$  just constructed cannot be assembled from exact finite-dimensional representations. In every finite group the doubling relation makes the image of  $\langle\gamma_0^2\rangle$  equal to the image of  $\langle\gamma_0\rangle$ , so the generator commutes with the required “halved” element and the distinguished word maps to the identity. A defining relator is already the identity *as an element* of the presented group, so containing it in the local set says nothing; what is needed is that the partial multiplication table evaluate the defining words and the distinguished word correctly. Take, then, a finite set containing the generators together with every intermediate prefix of every defining word and of the distinguished word. An exact local embedding of such a set satisfies the same equations. Hence neither  $E_{BS}$  nor its Clifford quotient is LEF, and the latter is not residually finite.



## S8. THE SCALING FAMILY

For  $m \geq 2$ , let  $E_m$  be the same eight-generator presentation, replacing the three relations  $tv_it^{-1} = v_i^2$  by  $tv_it^{-1} = v_i^m$  and leaving every other relator unchanged; thus  $E_2 = E$ . Write  $w_m$  for the image of the same distinguished word  $w$ .

**Corollary S8.1** (the scaling family). *For every natural number  $m \geq 2$ , the group  $E_m$  is finitely presented and  $w_m$  is a nontrivial central involution mapped to the identity by every homomorphism  $E_m \rightarrow \mathcal{U}(\mathcal{Q})$ . Consequently  $E_m$  is not MF, and neither  $C_{\max}^*(E_m)$  nor  $C_r^*(E_m)$  is an MF  $C^*$ -algebra.*

*Proof.* Replace the affine dilation  $\text{diag}(2, 2, 2, 1)$  by  $\text{diag}(m, m, m, 1)$  (both acting as in Lemma S2.2). It conjugates each integral translation  $v_i$  to  $v_i^m$  and fixes the three linear generators. Its inverse sends  $v_1$  to translation by  $e_1/m$ , which is not integral; hence the two Clifford generators remain distinct and  $w_m$  maps to  $-1$ . The Kazhdan base, the one-sided conjugation argument, and the central involution obstruction are independent of  $m$ .  $\square$

## S9. THE MF RESIDUAL

**Definition S9.1.** For a countable group  $G_1$ , the *MF residual* is

$$\text{Res}_{\text{MF}}(G_1) = \bigcap \{ \ker \Theta : \Theta \text{ a homomorphism } G_1 \rightarrow \mathcal{U}(\mathcal{Q}) \},$$

a normal subgroup of  $G_1$ . If  $G_1/N$  is MF then composing  $G_1 \rightarrow G_1/N$  with an injective homomorphism  $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$  gives a homomorphism with kernel  $N$ , so  $\text{Res}_{\text{MF}}(G_1) \leq N$ : it is the smallest normal subgroup with MF quotient, which is the sense in which *residual* is used here and for the nilpotent and finite residuals. The intersection is the same whether the dimension sequences are required to be strictly increasing or allowed to be arbitrary positive (Proposition 2.3). This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

**Lemma S9.2** (functoriality). *For every homomorphism  $\varphi: G_1 \rightarrow G_2$  of countable groups,*

$$\varphi(\text{Res}_{\text{MF}}(G_1)) \subseteq \text{Res}_{\text{MF}}(G_2).$$

*In particular, if  $\varphi: E \rightarrow G_2$  satisfies  $\varphi(w) \neq 1$ , then  $G_2$  is not MF.*

*Proof.* If  $x \in \text{Res}_{\text{MF}}(G_1)$  and  $\Theta$  is a homomorphism  $G_2 \rightarrow \mathcal{U}(\mathcal{Q})$ , then  $\Theta \circ \varphi$  is a homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ , so  $\Theta(\varphi(x)) = 1$ . For the second statement,  $w \in \text{Res}_{\text{MF}}(E)$  by Theorem A, so  $1 \neq \varphi(w) \in \text{Res}_{\text{MF}}(G_2)$ , and an injective homomorphism  $G_2 \rightarrow \mathcal{U}(\mathcal{Q})$  would map  $\varphi(w)$  to 1, a contradiction.  $\square$

**Proposition S9.3** (the universal MF quotient). *Let  $G_1$  be a countable group and  $R = \text{Res}_{\text{MF}}(G_1)$ .*

- (1)  *$R$  is the kernel of a single homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ .*
- (2)  *$G_1/R$  is MF.*



(3) Every homomorphism from  $G_1$  to an MF group factors through  $G_1/R$ . In particular  $G_1$  is MF if and only if  $R = \{1\}$ .

*Proof.* (3) If  $\varphi: G_1 \rightarrow M$  with  $M$  MF and  $j: M \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$  is an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$ , then  $j \circ \varphi$  is a homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ , so  $j(\varphi(R)) = 1$  and, by injectivity of  $j$ ,  $\varphi(R) = 1$ .

(1) and (2). Work on  $Q = G_1/R$ . Every homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$  maps  $R$  to 1, and for each  $g \notin R$  there is one with  $\Theta(g) \neq 1$ , which descends; so for every  $\bar{g} \neq 1$  in  $Q$  there is a homomorphism  $\Theta_{\bar{g}}: Q \rightarrow \mathcal{U}(\mathcal{Q})$  with  $\Theta_{\bar{g}}(\bar{g}) \neq 1$ . Enumerate the nonidentity elements and these representations, pass to coordinate unitary lifts, and diagonalize: at stage  $k$ , select in the  $i$ th of them, for each  $i \leq k$ , a coordinate at which multiplicativity holds within  $1/k$  on the first  $k$  group elements and at which *its own* designated element  $\bar{g}_i$  stays separated by its fixed positive constant — possible because  $\Theta_{\bar{g}_i}$  separates  $\bar{g}_i$  along infinitely many coordinates, which is all the hypothesis gives — and take the block sum. Block operator norms are maxima, so the single block coming from  $\Theta_{\bar{g}_i}$  already separates  $\bar{g}_i$ , and the sum therefore separates all of  $\bar{g}_1, \dots, \bar{g}_k$  at once. The resulting sequence is an operator-norm asymptotic representation separating every nonidentity element from the identity; its class, composed with the polar-correction isomorphism of Lemma 2.2, is a faithful homomorphism  $Q \rightarrow \mathcal{U}(\mathcal{Q})$ . This proves (2), and composing with  $G_1 \twoheadrightarrow Q$  proves (1). Finally, if  $G_1$  is MF then an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$  gives  $R = \{1\}$ ; if  $R = \{1\}$  then (2) says  $G_1$  is MF.  $\square$

**Corollary S9.4** (exact residual from a candidate quotient). *Let  $N \trianglelefteq G_1$  satisfy  $N \leq \text{Res}_{\text{MF}}(G_1)$ . Every homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$  factors uniquely through  $G_1/N$ . If in addition  $G_1/N$  is MF, then*

$$\text{Res}_{\text{MF}}(G_1) = N.$$

*Proof.* The containment  $N \leq \ker \Theta$  gives the unique quotient factorization for every such  $\Theta$ . For the reverse containment in the displayed equality, compose the quotient map  $G_1 \rightarrow G_1/N$  with an injective homomorphism  $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$ : its kernel is exactly  $N$ , while the MF residual lies in the kernel of every homomorphism to  $\mathcal{U}(\mathcal{Q})$ .  $\square$

**Corollary S9.5** (residual reduction to the quotient). *Let  $N \trianglelefteq G_1$  satisfy  $N \leq \text{Res}_{\text{MF}}(G_1)$  and let  $q: G_1 \rightarrow G_1/N$  be the quotient map. Then*

$$\text{Res}_{\text{MF}}(G_1) = q^{-1}(\text{Res}_{\text{MF}}(G_1/N)).$$

*Proof.* For the forward containment, Lemma S9.2 gives  $q(\text{Res}_{\text{MF}}(G_1)) \subseteq \text{Res}_{\text{MF}}(G_1/N)$ . Conversely, suppose  $q(x) \in \text{Res}_{\text{MF}}(G_1/N)$  and let  $\Theta$  be a homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ . Every homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$  maps  $N$  to 1 by hypothesis, so  $\Theta$  factors as  $\bar{\Theta} \circ q$  with  $\bar{\Theta}$  a homomorphism  $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$ ; hence  $\Theta(x) = \bar{\Theta}(q(x)) = 1$ .  $\square$

**Corollary S9.6.** *Let  $A$  be a unital  $C^*$ -algebra admitting an injective, possibly nonunital,  $*$ -homomorphism into a matrix quotient. Then the group  $E$  admits no injective homomorphism into  $\mathcal{U}(A)$ .*

*Proof.* Let  $j: A \rightarrow \mathcal{Q}$  be the possibly nonunital inclusion into the ambient matrix quotient and let  $\phi: E \rightarrow \mathcal{U}(A)$ . Complement correction gives

$$\hat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(\mathcal{Q}).$$

This is an injective homomorphism on  $\mathcal{U}(A)$ , so  $\hat{j} \circ \phi$  is a homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$ . By Theorem A,  $(\hat{j} \circ \phi)(w) = 1$ ; injectivity of  $\hat{j}$  gives  $\phi(w) = 1$ . Proposition S2.3 has  $w \neq 1$ , so  $\phi$  cannot be injective.  $\square$

### S9.1. Permanence and its failure.

**Lemma S9.7** (permanence). *Let all groups be countable.*

- (1) *Every subgroup of an MF group is MF.*
- (2) *Every residually finite group is MF.*
- (3) *Every locally finite group is MF.*

*Proof.* (1) Restrict an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$ . (2) For a finite test set  $F$ , a finite quotient injective on  $F$ , composed with its left regular permutation action, is an exact finite-dimensional representation; distinct permutations differ in some column of a permutation matrix, giving operator-norm separation at least 1, and the local-to-global direction of Proposition 2.3 concludes. (3) A finite test set  $F$  lies in a finite subgroup  $K$ . The left regular representation  $\lambda_K: K \rightarrow \mathcal{U}(\mathbb{C}^K)$  is a finite-dimensional unitary representation of  $K$  by permutation matrices. A local model is a map defined on all of  $G$ , so set

$$V(g) = \lambda_K(g) \quad (g \in K), \quad V(g) = 1 \quad (g \notin K).$$

On the test set  $V$  is exactly multiplicative, since a product of test elements again lies in  $K$  and  $\lambda_K$  is a homomorphism there; the values outside  $K$  are never used in the two conditions, which quantify over  $F$  only. It separates as in (2).  $\square$

**Corollary S9.8** (quotients). *MF groups are not closed under quotients: the free group  $F_8$  is MF, and the explicit group  $E$  is its quotient but is not MF.*

*Proof.* Map the eight free generators onto the eight displayed generators of  $E$ . The resulting homomorphism is surjective because those generators define the displayed presentation. The group  $F_8$  is residually finite, hence MF by Lemma S9.7(2), while  $E$  is not MF by Theorem A.  $\square$

## S10. ORBIT COLLAPSE

The argument of Theorem 4.1 extends from the negative spectral corner of a central involution to an arbitrary projection of  $\mathcal{Q}$  with commuting conjugation orbit.

**Theorem S10.1** (projection collapse). *Let  $H$  be countable, let  $L \leq H$  have property (T), let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ , and let  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  be a homomorphism. Let  $p \in \mathcal{Q}$  be a projection such that  $p$  commutes with  $\Theta(s\gamma s^{-1})$  for every  $\gamma \in L$  and the projections  $\Theta(\gamma)p\Theta(\gamma)^*$ ,  $\gamma \in L$ , commute pairwise. Then  $p$  commutes with  $\Theta(L)$ .*

*Proof.* Property (T) makes  $L$  finitely generated; fix a finite symmetric generating set  $S = \{a_1, \dots, a_m\}$ , so that  $S = S^{-1}$  and every element of  $L$  is a product of elements of  $S$  with no inverses taken. Write  $p_g = \Theta(g)p\Theta(g)^*$  and  $d_g = p_g - p$  for  $g \in L$ , and suppose  $d_{g_0} \neq 0$  for some  $g_0 \in L$ .

Since  $p_g$  and  $p$  are commuting projections,  $d_g$  is self-adjoint with  $d_g^3 = d_g$ , so  $d_g^2$  is a projection and  $d_g d_g^2 = d_g$ ; write  $q_g = d_g^2$ . All the  $p_g$  commute, hence so do all the  $d_g$  and all the  $q_g$ . Put

$$q = 1 - \prod_{a \in S} (1 - q_a),$$

the join of the projections  $q_a$ ,  $a \in S$ . The elements  $g$  with  $p_g = p$  form a subgroup of  $L$ , so if  $q = 0$ , and hence  $d_a = 0$  for every  $a \in S$ , then  $d_g = 0$  for every  $g \in L$ , against  $d_{g_0} \neq 0$ . Thus  $q \neq 0$ .

A Hilbert space attached to  $q$ . Take a self-adjoint bounded lift  $(x_n)$  of  $q$ . Then  $\|x_n^2 - x_n\| \rightarrow 0$ , so the spectrum of  $x_n$  clusters at  $\{0, 1\}$  and continuous functional calculus makes  $Q_n = \chi_{[1/2, \infty)}(x_n)$  a projection with  $\|Q_n - x_n\| \rightarrow 0$ ; the sequence  $(Q_n)$  again lifts  $q$ . Since  $q \neq 0$ ,  $Q_n \neq 0$  for infinitely many  $n$ ; discard the remaining coordinates, relabel by  $\mathbb{N}$ , and put  $k_n = \text{rank } Q_n \geq 1$ .

Give  $M_{d_n}(\mathbb{C})$  the inner product  $\langle x, y \rangle_n = \text{Tr}(y^*x)/k_n$ , write  $\|x\|_F = \text{Tr}(x^*x)^{1/2}$  for the unnormalized Hilbert–Schmidt norm, and let  $K_\omega$  be the Hilbert-space ultraproduct of these spaces along a free ultrafilter  $\omega$ . Let  $\mathcal{I}_q$  be the set of finite sums  $\sum_{j \leq r} x_j q y_j$  with  $x_j, y_j \in \mathcal{Q}$ , a two-sided ideal of  $\mathcal{Q}$  invariant under inner automorphisms.

Given such a sum, choose bounded coordinate lifts  $X_{j,n}$  of  $x_j$  and  $Y_{j,n}$  of  $y_j$  and put  $Z_n = \sum_{j \leq r} X_{j,n} Q_n Y_{j,n}$ , of rank at most  $r k_n$ . The square of the Frobenius norm is the sum of the squares of the singular values, of which at most  $\text{rank}(Z_n)$  are nonzero and each is at most the operator norm, so

$$(11) \quad \|Z_n\|_F^2 \leq \text{rank}(Z_n) \|Z_n\|^2 \leq r k_n \|Z_n\|^2$$

and  $(Z_n)$  is a bounded sequence for the inner products  $\langle \cdot, \cdot \rangle_n$ . Its class in  $K_\omega$  depends only on  $z = \sum_j x_j q y_j$ : a sequence  $Z'_n$  obtained from a second decomposition, with  $r'$  summands, satisfies  $\|Z_n - Z'_n\| \rightarrow 0$  and  $\text{rank}(Z_n - Z'_n) \leq (r + r')k_n$ , so  $\|Z_n - Z'_n\|_F^2/k_n \rightarrow 0$  by (11). Write  $\Lambda(z) = [Z_n]_\omega$ . The map  $\Lambda: \mathcal{I}_q \rightarrow K_\omega$  is linear,

$$\|\Lambda(q)\| = \lim_\omega \left( \frac{\text{Tr } Q_n}{k_n} \right)^{1/2} = 1,$$

and  $\Lambda(z) = 0$  implies  $\Lambda(xzy) = 0$  for all  $x, y \in \mathcal{Q}$ , since bounded lifts of  $x$  and  $y$  multiply  $\|Z_n\|_F$  by a bounded factor.

*An exact cocycle.* For  $a \in S$  the projection  $q_a$  annihilates  $\prod_{b \in S} (1 - q_b)$ , so  $q_a q = q_a$  and  $d_a = d_a q_a = d_a q \in \mathcal{I}_q$ . From  $p_{gh} = \Theta(g) p_h \Theta(g)^*$ ,

$$(12) \quad d_{gh} = d_g + \Theta(g) d_h \Theta(g)^* \quad (g, h \in L),$$

and  $\mathcal{I}_q$  is inner-invariant, so  $d_g \in \mathcal{I}_q$  for every  $g \in L$ .

Choose unitary coordinate lifts  $U_n(h)$  of  $\Theta(h)$  (Lemma 2.2). Conjugation by a unitary preserves each  $\langle \cdot, \cdot \rangle_n$ , and  $\|\text{Ad } u - \text{Ad } v\| \leq 2 \|u - v\|$ , so  $\pi(h) = [\text{Ad } U_n(h)]_\omega$  defines a unitary representation of  $H$  on  $K_\omega$ , and  $\pi(h)\Lambda(z) = \Lambda(\Theta(h)z\Theta(h)^*)$  on the closed invariant subspace  $K_q = \overline{\Lambda(\mathcal{I}_q)}$ . By (12) the map  $\beta(g) = \Lambda(d_g)$  satisfies

$$\beta(gh) = \beta(g) + \pi(g)\beta(h) \quad (g, h \in L),$$

a 1-cocycle for  $\pi|_L$  with values in  $K_q$ .

Suppose  $\beta(a) = 0$  for every  $a \in S$ . Put  $r_1 = 1$  and  $r_i = \prod_{j < i} (1 - q_{a_j})$  for  $i \geq 2$ . The products  $e_i = r_i q_{a_i}$  are projections with  $e_i = r_i - r_{i+1}$ , so they are pairwise orthogonal and  $\sum_i e_i = 1 - r_{m+1} = q$ . Each  $d_{a_i}$  commutes with  $r_i$ , so  $e_i = (d_{a_i} r_i) d_{a_i}$  and  $\Lambda(e_i) = 0$ , whence  $\Lambda(q) = 0$  against  $\|\Lambda(q)\| = 1$ . So  $\beta$  is not identically zero.

*Property (T).* By the Delorme–Guichardet theorem [BHV, Theorem 2.12.4],  $\beta$  is a coboundary: there is  $y \in K_q$  with  $\beta(g) = y - \pi(g)y$  for every  $g \in L$ . The hypothesis on  $p$  gives  $d_{sas^{-1}} = 0$ , hence  $\pi(sas^{-1})y = y$ , for every  $a \in L$ . The proof of Theorem 3.4 identifies  $\text{Fix } \pi(sLs^{-1})$  with  $\text{Fix } \pi(L)$ : the projection  $P$  onto  $\text{Fix } \pi(L)$  and its conjugate  $Q = \pi(s)P\pi(s)^*$ , whose range is  $\text{Fix } \pi(sLs^{-1})$ , satisfy  $P \leq Q$  and  $P \sim Q$  in the norm ultraproduct, which is finite, so  $P = Q$  by Lemma 3.1. The weight  $k_n$  in place of  $d_n$  leaves that comparison unchanged (Theorem S4.1(1)). Hence  $\pi(g)y = y$  and  $\beta(g) = 0$  for every  $g \in L$ , contradicting the previous paragraph. So  $d_g = 0$ , that is,  $p$  commutes with  $\Theta(g)$ , for every  $g \in L$ .  $\square$

Equivalently, every projection of  $\Theta(sLs^{-1})' \cap \mathcal{Q}$  whose  $\Theta(L)$ -conjugates commute pairwise already lies in  $\Theta(L)' \cap \mathcal{Q}$ .

**Definition S10.2** (involutive orbit witness). Let  $H$  be a group,  $L \leq H$  a subgroup, and  $s \in H$ . An element  $k \in H$  is an *involutive orbit witness* for  $(L, s)$  if

- (W1)  $k^2 = 1$ ;
- (W2)  $s\gamma s^{-1}$  commutes with  $k$  for every  $\gamma \in L$ ;
- (W3) the conjugates  $\gamma_1 k \gamma_1^{-1}$  and  $\gamma_2 k \gamma_2^{-1}$  commute for all  $\gamma_1, \gamma_2 \in L$ .

The pair  $(L, s)$  determines the normal subgroup

$$D_{\text{coll}}(L, s) = \langle\langle [\gamma, k] : \gamma \in L, k \text{ a witness for } (L, s) \rangle\rangle_H.$$

In a permutational wreath product  $K^{(X)} \rtimes G$  with  $L \leq G$  conjugated into itself by  $s$  and  $x_0 \in X$  a point fixed by  $sLs^{-1}$ , every involution of the copy of  $K$  at  $x_0$  is a witness: copies at distinct points commute, and the conjugate copy fixes the point.

**Theorem S10.3** (involutive collapse). *Let  $H$  be countable, let  $L \leq H$  be a subgroup with property (T), and let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ . Then*

$$D_{\text{coll}}(L, s) \leq \text{Res}_{\text{MF}}(H).$$

*Proof.* Let  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  be a homomorphism and  $k$  a witness for  $(L, s)$ . By (W1) the unitary  $u = \Theta(k)$  is self-adjoint, so  $p = \frac{1}{2}(1 - u)$  is a projection, and

$$\Theta(\gamma)p\Theta(\gamma)^* = \frac{1}{2}(1 - \Theta(\gamma k \gamma^{-1})) \quad (\gamma \in L).$$

By (W2) the projection  $p$  commutes with  $\Theta(s\gamma s^{-1})$  for  $\gamma \in L$ , and by (W3) the displayed projections commute pairwise. Theorem S10.1 makes  $p$ , and hence  $u = 1 - 2p$ , commute with  $\Theta(L)$ , so  $\Theta([\gamma, k]) = 1$  for every  $\gamma \in L$ . Kernels are normal, so  $\Theta$  is trivial on  $D_{\text{coll}}(L, s)$ .  $\square$

**Theorem S10.4** (torsion collapse). *Let  $H$  be countable, let  $L \leq H$  have property (T), let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ , and let  $k \in H$  have finite order. If the conjugate copy  $sLs^{-1}$  centralizes  $k$  and the conjugates  $\gamma k \gamma^{-1}$ ,  $\gamma \in L$ , commute pairwise, then every homomorphism  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta([\gamma, k]) = 1$  for every  $\gamma \in L$ ; the normal closure of all such commutators, over all finite-order witnesses for  $(L, s)$ , lies in  $\text{Res}_{\text{MF}}(H)$ , with equality whenever the corresponding quotient of  $H$  is MF.*

*Proof.* Let  $\Theta$  be a homomorphism to  $\mathcal{U}(\mathcal{Q})$ ,  $m$  the order of  $k$ , and  $u = \Theta(k)$ , a unitary with  $u^m = 1$ . Its discrete Fourier idempotents  $p_j = m^{-1} \sum_{t \in \mathbf{Z}/m} e^{-2\pi i j t / m} u^t$ , for  $j \in \mathbf{Z}/m$ , are self-adjoint idempotents of  $\mathcal{Q}$ . Each  $p_j$  is a polynomial in  $u$ , so whatever commutes with  $u$  commutes with every  $p_j$ : the represented conjugate copy centralizes each  $p_j$ , and the  $\Theta(L)$ -conjugates of each  $p_j$  commute pairwise because the conjugated witnesses do. Theorem S10.1 therefore makes every  $p_j$  commute with  $\Theta(L)$ , and Fourier reconstruction expresses  $u = \sum_j e^{2\pi i j / m} p_j$ , so  $u$  commutes with  $\Theta(L)$  and  $\Theta([\gamma, k]) = 1$  for every  $\gamma \in L$ .  $\square$

*Proof of Theorem B.* Write  $q: E \rightarrow E/\langle w \rangle$ ,  $\bar{d} = q(d)$ , and  $\bar{L} = q(\iota(\mathcal{B}))$ . The element  $\bar{d}$  is an involution because  $c^2 = 1$ , and  $q(t)\bar{L}q(t)^{-1}$  centralizes  $\bar{d}$  because  $t\iota(\mathcal{B})t^{-1}$  commutes with  $c$  in  $E$ . Property (T) passes to  $\bar{L}$ , and  $q(t)$  conjugates it into itself. Conditions (W1) and (W2) of Definition S10.2 therefore hold. It remains to verify (W3).

The linear generators  $x, y, z$  commute with  $d$ , and  $t$  carries each  $v_i$  to  $v_i^2$ , so  $\iota(v_i^2)$  commutes with  $d$  as well. Hence  $\iota(g)d\iota(g)^{-1}$  depends only on the exponent vector of the translation part of  $g$  modulo 2, which leaves eight conjugates, indexed by  $(\mathbb{Z}/2)^3$ . By Theorem 5.1 the distinguished word is  $w = [d, \iota(v_1)d\iota(v_1)^{-1}]$ , the commutator attached to the class of  $v_1$ . The six words  $x, x^2, z, yz, yx, y^2x$  in the linear generators carry that class onto the six remaining nonzero classes, and each commutes with  $d$ ; conjugating the displayed identity by the corresponding  $\iota(r)$  exhibits every nonzero parity commutator as a conjugate of  $w$ , while the zero class contributes  $[d, d] = 1$ .

Since  $w$  is central, all eight lie in  $\langle w \rangle$ , and one further conjugation reduces an arbitrary pair of conjugates to this case. Thus  $\bar{d}$  is an involutive orbit witness for  $(\bar{L}, q(t))$ , and by Theorem S10.3  $[q(\iota(v_1)), \bar{d}] \in \text{Res}_{\text{MF}}(E/\langle w \rangle)$ .

Nontriviality passes through the homomorphism  $E \rightarrow W$  of Proposition S2.3: it carries  $[\iota(v_1), d]$  to the product of the Clifford generators at the two distinct points  $\tau o$  and  $j(\bar{v}_1)\tau o$ , and carries  $\langle w \rangle$  into  $\langle \zeta \rangle$ , whose configuration coordinate is zero. Hence  $[\iota(v_1), d] \notin \langle w \rangle$ , its image in  $E/\langle w \rangle$  is nontrivial, and Proposition S9.3 makes  $E/\langle w \rangle$  non-MF. Corollary S9.5 identifies  $\text{Res}_{\text{MF}}(E)$  with the  $q$ -preimage of  $\text{Res}_{\text{MF}}(E/\langle w \rangle)$ , so  $u$  belongs to it.  $\square$

**Corollary S10.5** (collapse computes the residual). *In the setting of Theorem S10.3, write  $q: H \rightarrow H/D_{\text{coll}}(L, s)$  for the quotient map. Then*

$$\text{Res}_{\text{MF}}(H) = q^{-1}(\text{Res}_{\text{MF}}(H/D_{\text{coll}}(L, s))),$$

*and if  $H/D_{\text{coll}}(L, s)$  is MF, then  $\text{Res}_{\text{MF}}(H) = D_{\text{coll}}(L, s)$ .*

*Proof.* Combine Theorem S10.3 with Corollaries S9.5 and S9.4.  $\square$

In the second case  $H/D_{\text{coll}}(L, s)$  is the universal MF quotient of  $H$  (Proposition S9.3): every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  factors through it, and it is MF.

## S11. GROUP ALGEBRAS, SOFICITY, AND PERMANENCE

**S11.1. The reduced group algebra of  $E$ .** The stable finiteness in Theorem C comes from two standard facts.

**Lemma S11.1.** *Let  $G_1$  be a countable discrete group.*

- (1) *The canonical tracial state  $\tau$  on  $C_r^*(G_1)$  is faithful.*
- (2) *Every unital  $C^*$ -algebra  $A$  with a faithful tracial state is stably finite: for every  $k \geq 1$ , every isometry in  $M_k(A)$  is a unitary.*

*Proof.* (1) Realize  $C_r^*(G_1) \subseteq B(\ell^2 G_1)$  as the norm closure of the span of the left translation unitaries  $\lambda_g$ . Let  $\rho_g$  be the right translation unitaries,  $(\rho_g \xi)(h) = \xi(hg)$ . Each  $\rho_g$  commutes with each  $\lambda_{g'}$ , hence with every element of  $C_r^*(G_1)$ . The canonical trace is  $\tau(x) = \langle x\delta_e, \delta_e \rangle$ . If  $\tau(x^*x) = 0$  then  $\|x\delta_e\| = 0$ , so  $x\delta_e = 0$ , and then  $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$  for every  $g \in G_1$ . The vectors  $\delta_g$  are total in  $\ell^2 G_1$ , so  $x = 0$ .

(2) On  $M_k(A)$  the functional  $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$  is a state with the trace property  $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$ , and it is faithful: the  $i$ -th diagonal entry of  $x^*x$  is  $\sum_j x_{ji}^*x_{ji}$ , so  $\tau_k(x^*x) = 0$  implies  $\tau(x_{ji}^*x_{ji}) = 0$ , hence  $x_{ji} = 0$ , for all  $i, j$ . If  $v \in M_k(A)$  satisfies  $v^*v = 1$ , then  $y := 1 - vv^*$  is a projection with  $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$ ; since  $y = y^*y$ , faithfulness gives  $y = 0$ , i.e.  $vv^* = 1$ .  $\square$

**S11.2. The Clifford quotient and permanence.** The map  $E \rightarrow W$  of Proposition S2.3 is surjective: its image contains  $\Sigma$  and the generator  $c_o$  at the base point, hence every Clifford generator by transitivity of the  $\Sigma$ -action on  $X$ . Since  $E$  is generated by  $v_1, x, y, z, t, c$  (Section S1), their six images generate  $W$ . The map sends  $w \in \text{Res}_{\text{MF}}(E)$  to the nontrivial sign  $\zeta$ ; functoriality (Lemma S9.2) makes  $\zeta \in \text{Res}_{\text{MF}}(W)$ . Hence the finitely generated group  $W$  is not MF, and neither is  $C_r^*(W)$ : an MF embedding of it would restrict to a faithful homomorphism  $W \rightarrow \mathcal{U}(\mathcal{Q})$  on the canonical group unitaries.

Failure of extension permanence. MF groups are not closed under extensions. More precisely,  $W$  fits into an exact sequence

$$1 \longrightarrow \text{Cl}(X) \longrightarrow W \longrightarrow \Sigma \longrightarrow 1$$

whose kernel is locally finite — hence LEF, sofic and MF — and whose quotient is MF, while  $W$  itself is finitely generated and not MF. Nearby approximation properties do have extension permanence of this shape: an extension of a residually finite group by a residually finite group is weakly sofic [Gl], and weak soficity is detected by finite targets [GR].

*Proof.* By Lemma S9.7(3), the Clifford group is locally finite, hence MF. The group  $\Sigma$  identifies with the subgroup of  $\text{GL}_4(\mathbb{Q})$  generated by  $\bar{\Gamma}$  and  $D$ : the levels are  $D^{-n}\bar{\Gamma}D^n$ , and the determinant gives the  $\mathbb{Z}$ -coordinate. It is residually finite. At every odd modulus the determinant 8 of the dilation is invertible, so the congruence reduction of the integral affine model descends to conjugation by the dilation and extends over the direct limit and its shift. These reductions separate any nontrivial element; compare Mal'cev's theorem [Mal] for finitely generated linear groups. Thus  $\Sigma$  is MF by Lemma S9.7(2), whereas  $W$  is not MF by the preceding construction.  $\square$

Extensions by  $\mathbb{Z}$ . Composing the two canonical projections gives a surjection  $W \rightarrow \mathbb{Z}$  whose kernel is  $\text{Cl}(X) \rtimes T$ , and this kernel is LEF, hence MF: every finite subset lies in  $L \rtimes \bar{\Gamma}_n$  for a finite invariant subgroup  $L \leq \text{Cl}(X)$  and one level, and that group is residually finite because the level acts on the finite group  $L$  through a finite quotient. Thus  $W$  is an extension of an LEF group by  $\mathbb{Z}$  and is not MF, while soficity is preserved by every extension with amenable quotient, as in Elek–Szabó [ES06].

A simple sofic envelope. There is a countable simple sofic group  $S_{\text{simp}}$  with

$$\text{Res}_{\text{MF}}(S_{\text{simp}}) = S_{\text{simp}}.$$

Equivalently, every homomorphism from  $S_{\text{simp}}$  to an MF group is trivial.

*Proof.* By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed  $E$ , which is sofic by Theorem D, in such a group. Functoriality of the MF residual sends  $w$  to a nontrivial element of the MF residual of the simple group. That residual is normal, so it is the whole group. Every homomorphism to an MF group maps the MF residual to 1.  $\square$



**S11.3. Soficity of the literal group.** Deleting the letter  $c$  and the families (C) and (W) from (9) leaves the presentation  $\langle \mathcal{B}, t \mid tut^{-1} = \bar{\alpha}(u) \rangle$  of the ascending HNN extension of the base along the doubling endomorphism. By Remark S1.2 the base is  $\bar{\Gamma}$ , so

$$E/\langle\langle c \rangle\rangle \cong \Sigma$$

for the group  $\Sigma = T \rtimes \mathbb{Z}$  of Section S2, with its site set  $X = \Sigma/B$ , base point  $o$ , shift  $\tau$  and level-zero base  $B = j(\bar{\Gamma})$ . The levels

$$\bar{\Gamma}_n = \tau^{-n} B \tau^n \quad (n \geq 0)$$

form an ascending chain of copies of  $\bar{\Gamma}$  with union  $T$ . Put  $K = \tau^{-1} B \tau$ ; then  $B \leq K$  and  $[K : B] = [\bar{\Gamma} : \bar{\alpha}(\bar{\Gamma})]$ , at most eight. The cosets  $\Sigma/K$  are the *blocks*, and each block, as a set of sites, has  $[K : B]$  elements.

Let  $\mathcal{G}$  be the graph on  $X$  whose edges are the  $\Sigma$ -orbit of the pair  $\{\tau o, j(\bar{v}_1)\tau o\}$ , whose entries are distinct by Lemma S2.2. Two sites are adjacent in  $\mathcal{G}$  exactly when they are distinct and lie in a common block. Adjacent sites share a block because  $(\tau o)^{-1}(j(\bar{v}_1)\tau o) = \tau^{-1}j(\bar{v}_1)\tau \in K$ . Conversely, the chart  $u\bar{\alpha}(\bar{\Gamma}) \mapsto \tau^{-1}j(u)\tau o$  identifies the block of  $o$  with  $\bar{\Gamma}/\bar{\alpha}(\bar{\Gamma})$  and the generating edge with the pair  $\{\bar{\alpha}(\bar{\Gamma}), \bar{v}_1\bar{\alpha}(\bar{\Gamma})\}$ ; since  $\bar{\alpha}(\bar{\Gamma})$  acts transitively on the nontrivial cosets of  $\bar{\alpha}(\bar{\Gamma})$ , every pair of distinct sites of a block is the image of the generating pair.

For a graph  $\mathcal{G}$  on  $X$ , let  $C(\mathcal{G})$  be the group presented by generators  $\zeta$  and  $c_\xi$  ( $\xi \in X$ ) and relations

$$\zeta^2 = c_\xi^2 = 1, \quad [\zeta, c_\xi] = 1, \quad [c_\xi, c_\eta] = \zeta \text{ for each edge } \{\xi, \eta\} \text{ of } \mathcal{G}.$$

For the complete graph on  $X$  this is  $\text{Cl}(X)$  of Construction S2.1, and adjoining the anticommutation relations at the remaining pairs gives a surjection  $C(\mathcal{G}) \rightarrow \text{Cl}(X)$ .

**Proposition S11.2** (normal form for  $E$ ). *There is an isomorphism*

$$E \cong C(\mathcal{G}) \rtimes \Sigma$$

*under which  $\langle\langle c \rangle\rangle$  is the factor  $C(\mathcal{G})$ , the conjugate  $gcg^{-1}$  is the lamp  $c_{gB}$ , and the distinguished word  $w$  is the sign  $\zeta$ . Each level  $\bar{\Gamma}_n$  has finite orbits on  $X$ .*

*Proof.* Send the six base letters and  $t$  to the corresponding generators of  $\Sigma$ , and  $c$  to  $c_o$ . Families (B) and (T) of (9) hold in  $\Sigma$  by construction. For family (C):  $c_o^2 = 1$ , and  $B$  fixes  $o$ , so each base generator fixes  $c_o$ . For family (W): the pair  $\{\tau o, j(\bar{v}_1)\tau o\}$  is an edge, so the image of  $w$  is  $[c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = \zeta$ , which is central in  $C(\mathcal{G})$  and fixed by  $\Sigma$ . This defines  $E \rightarrow C(\mathcal{G}) \rtimes \Sigma$ .

In the other direction,  $\zeta \mapsto w$  and  $c_{gB} \mapsto gcg^{-1}$  respects the defining relations of  $C(\mathcal{G})$ : the assignment is independent of the coset representative because the base centralizes  $c$ ; the images are involutions;  $w$  is central of order two; and along an edge  $g\{\tau o, j(\bar{v}_1)\tau o\}$  the commutator of the two



images is  $gw g^{-1} = w$ ; the presentation imposes relations only along edges. Combined with the splitting  $\Sigma \rightarrow E$  on the seven remaining letters, this gives  $C(\mathcal{G}) \rtimes \Sigma \rightarrow E$ . The two composites fix the eight letters of  $E$  and the generators  $\zeta, c_\xi$  of  $C(\mathcal{G})$ , hence are the identity.

For the orbits,  $\bar{\alpha}(\bar{\Gamma})$  has index at most eight in  $\bar{\Gamma}$  — the parity classes of the translation lattice — so  $\tau$  conjugates  $B$  onto a subgroup of finite index in itself. Every  $\bar{\Gamma}_n$  is therefore commensurable with  $B$ , all of  $\Sigma$  commensurates  $B$ , and orbit-stabilizer makes each  $\bar{\Gamma}_n$ -orbit in  $X = \Sigma/B$  finite.  $\square$

Composing the isomorphism with  $C(\mathcal{G}) \rightarrow \text{Cl}(X)$  gives the surjection  $E \rightarrow W$  of Proposition S2.3. The group  $\text{Cl}(X)$  is locally finite;  $C(\mathcal{G})$  is not, since two lamps at sites of different blocks generate an infinite dihedral group and three blocks give a nonabelian free subgroup. For a finite set  $S$  of sites write

$$M_S = \langle \zeta, c_\xi \ (\xi \in S) \rangle \leq C(\mathcal{G}).$$

**Lemma S11.3** (windows are residually finite). *For every finite set  $S$  of sites the subgroup  $M_S$  is residually finite.*

*Proof.* Killing  $\zeta$  sends  $C(\mathcal{G})$  onto the free product of the elementary abelian 2-groups of the blocks, and  $M_S$  into the free product of the finitely many factors at blocks meeting  $S$ . A free product of finite groups is residually finite. The kernel of  $C(\mathcal{G}) \rightarrow C(\mathcal{G})/\langle\langle \zeta \rangle\rangle$  is the normal closure of  $\zeta$ , and  $\zeta$  is central, so that normal closure is  $\langle \zeta \rangle$ ; hence the kernel of this map on  $M_S$  lies in  $\langle \zeta \rangle$ . (No splitting is available or needed: a two-lamp block is the Clifford extension  $D_8 \rightarrow (\mathbb{Z}/2)^2$ , which does not split over the whole factor.) The surjection  $C(\mathcal{G}) \rightarrow \text{Cl}(X)$  sends  $\zeta$  to the sign of  $\text{Cl}(X)$ , which is nontrivial, and  $\text{Cl}(X)$  is locally finite, so the image of the finitely generated group  $M_S$  there is finite. An element of  $M_S$  in the kernel of the first map lies in  $\langle \zeta \rangle$ , and the second map separates the two elements of that subgroup.  $\square$

*Proof of Theorem D.* Let  $E_0$  be the kernel of the homomorphism  $E \rightarrow \mathbb{Z}$  given by the exponent sum in  $t$ . By Proposition S11.2,  $E_0 \cong C(\mathcal{G}) \rtimes T$ .

Take a finite subset of  $E_0$ . Its  $T$ -coordinates lie in one level  $\bar{\Gamma}_n$ , the chain being ascending, and its  $C(\mathcal{G})$ -coordinates lie in  $M_{S_0}$  for some finite set  $S_0$  of sites, each generator of  $C(\mathcal{G})$  being a single lamp or the sign. Let  $S$  be the union of the  $\bar{\Gamma}_n$ -orbits of the sites in  $S_0$ , a finite set by Proposition S11.2. Then  $M_S$  is  $\bar{\Gamma}_n$ -invariant and contains  $M_{S_0}$ , so the finite subset lies in  $M_S \rtimes \bar{\Gamma}_n$ .

That group is residually finite. Indeed  $M_S$  is residually finite by Lemma S11.3, the level  $\bar{\Gamma}_n \cong \bar{\Gamma}$  is residually finite through the congruence quotients of its matrix model (Lemma S2.2), and  $\bar{\Gamma}_n$  acts on  $M_S$  through permutations of the finite set  $S$ , hence through a finite quotient  $F$ . The preimage of the identity of  $F$  is a subgroup of finite index isomorphic to  $M_S \times \ker(\bar{\Gamma}_n \rightarrow F)$ , which is residually finite, and residual finiteness passes to a group containing a residually finite subgroup of finite index.

So  $E_0$  is a directed union of subgroups embedding in residually finite groups, that is, LEF, and therefore sofic. Now  $E$  is a split extension of  $E_0$  by  $\mathbb{Z}$ , split by  $t$ , so  $E$  is sofic by the Elek–Szabó amenable-extension theorem [ES06], and hyperlinear because sofic groups are. Failure of MF is Theorem A. LEF groups are MF [CDE], so  $E$  is not LEF; a residually finite group is LEF, so  $E$  is not residually finite either.  $\square$

The same decomposition makes  $E$  exact. Killing  $\zeta$  sends  $C(\mathcal{G})$  onto the free product of the elementary abelian 2-groups of the blocks, each of order at most  $2^8$ , and free products, directed unions and extensions of exact groups are exact [KW2]; so that free product is exact, and  $C(\mathcal{G})$ , a central extension of it by  $\langle \zeta \rangle$ , is exact as well. The group  $\Sigma$  is a subgroup of  $\mathrm{GL}_4(\mathbb{Q})$  and therefore exact [GHW], and  $E \cong C(\mathcal{G}) \rtimes \Sigma$ . For a discrete group exactness is equivalent to exactness of the reduced group algebra [KW1], so  $C_r^*(E)$  is exact. The same applies to  $W$ , whose Clifford kernel is locally finite and hence amenable.

## S12. MARKED LIMITS, QUASI-IDENTITIES, AND UNDECIDABILITY

### S12.1. Marked-limit closure.

**Theorem S12.1** (the MF locus is closed). *Fix  $k \geq 1$  and consider the space of  $k$ -marked groups — quotients of the free group  $F_k$ , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators.*

- (1) *The marked groups with MF quotient form a closed set.*
- (2) *Equivalently, the non-MF marked groups form an open set.*
- (3) *Every non-MF  $k$ -marked group admits a finite-radius certificate: there is a radius  $R$  such that every  $k$ -marked group whose relations agree with it on the reduced words of length at most  $R$  is non-MF.*

*Proof.* A marked group fails MF exactly when normalized local approximation fails on some finite subset of it. Expressing the finitely many products and inequalities of that subset by words in the marked generators produces finitely many relation constraints, and every marked group satisfying those constraints has the same finite subset with no local model; so a basic clopen neighborhood fails MF. Enclosing the constraint words in a reduced-word ball gives the radius form.  $\square$

**S12.2. A finite universal Horn obstruction.** Every MF group satisfies a single finite quasi-identity: a universal implication between finitely many equations, that is, a universal Horn sentence. Write  $R$  for the forty-one relators of (9), regarded as words in eight variables, and  $w$  for the distinguished word in those variables.

**Proposition S12.2** (finite quasi-identity). *Every MF group  $G$  satisfies the finite universal Horn sentence*

$$\forall x_1, \dots, x_8: \bigwedge_{r \in R} r(x_1, \dots, x_8) = 1 \implies w(x_1, \dots, x_8) = 1,$$

while the canonical generating tuple of  $E$  satisfies every relator in  $R$  and has  $w \neq 1$ .

*Proof.* A tuple satisfying the relators induces a homomorphism  $E \rightarrow G$  sending the distinguished word of  $E$  to  $w(x_1, \dots, x_8)$ . If  $G$  is MF, that image is trivial:  $w \in \text{Res}_{\text{MF}}(E)$  by Theorem A, and the residual is functorial (Lemma S9.2). The canonical tuple of  $E$  satisfies  $R$  by definition, and its distinguished word is nontrivial by Theorem A.  $\square$

**Corollary S12.3** (a basic clopen set of non-MF marked groups). *In the space of eight-generator marked groups — quotients of the free group  $F_8$ , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators — the marked groups satisfying all relators of  $R$  and  $w \neq 1$  form a nonempty clopen set consisting entirely of non-MF groups.*

*Proof.* Both conditions constrain finitely many values of the marked multiplication, so the set is clopen; it contains the marked group  $E$  by Theorem A, and Proposition S12.2 excludes every MF point.  $\square$

**S12.3. Undecidability of MF recognition.** Lemma S9.7(1) and Theorem A make MF a Markov property of finitely presented groups: the trivial group is MF, and by subgroup heredity the finitely presented group  $E$  embeds in no MF group. The Adian–Rabin theorem [Ra] therefore applies, and classically MF recognition is undecidable.

Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write  $G_e$  for the group presented by the code  $e$ .

**Corollary S12.4** (undecidability, given a halting reduction). *Suppose given a computable map  $e$  sending each instance  $i$  of the halting problem to a presentation code  $e(i)$  such that  $G_{e(i)}$  is MF exactly when  $i$  halts. Then no algorithm decides, from a finite presentation, whether the presented group is MF, and the set of presentation codes of non-MF groups is not recursively enumerable.*

Effectiveness of the coding is necessary: a map attaching a finitely presented group to each code need not be computable, and an uncomputable choice would make the statement true for reasons having nothing to do with groups.

Constructing such a map is the Adian–Rabin argument [Ra] applied to the Markov property “is MF”. It fixes one finite presentation  $P_0$  whose group has undecidable word problem and takes a word  $v$  in the generators of  $P_0$

to a finite presentation whose group is trivial, hence MF, when  $v = 1$  in  $P_0$ , and contains  $E$ , hence non-MF by subgroup heredity, when  $v \neq 1$ . The words representing 1 form a recursively enumerable set that is not recursive, so the words with  $v \neq 1$  form a set that is not recursively enumerable, and an enumeration of the non-MF presentations would enumerate it. This paragraph is classical and rests on [Ra]; the verification records below cover subgroup heredity, the existence of a finitely presented non-MF group, and the undecidability statement *given* a reduction, which is the hypothesis of Corollary S12.4. The Adian–Rabin construction itself is not formalized, and no record above claims it. Finitely presented non-MF groups occur in infinite multiplicity, and finitely generated ones in continuum multiplicity. The groups  $E \times \mathbb{Z}^k$  for  $k \geq 0$  are finitely presented, pairwise nonisomorphic, and non-MF, each containing  $E$  (Lemma S9.7(1)). They are pairwise nonisomorphic because their abelianizations have distinct torsion-free ranks; the number of homomorphisms to  $\mathbb{Z}/2$  separates them as well. Beyond finite presentation, pairing  $E$  with a continuum of pairwise nonisomorphic finitely generated groups  $N$  — Neumann’s two-generator groups [Ne], say — yields  $2^{\aleph_0}$  pairwise nonisomorphic finitely generated non-MF groups: each product  $E \times N$  is non-MF, while a fixed countable group  $X$  satisfies  $X \cong E \times N$  for at most countably many of the isomorphism types  $N$ , since each such  $N$  embeds in  $X$  as a finitely generated subgroup and a countable group has only countably many of those.

### S13. LIMITATIONS OF THE OPERATOR-NORM METHOD

The proof is specific to the operator norm. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm: the estimate  $\|\text{Ad}(U) - \text{Ad}(V)\| \leq 2\|U - V\|$  has no converse, and there are unitaries of arbitrarily small normalized Hilbert–Schmidt distance whose adjoints are at operator-norm distance 2. The proof of Theorem 3.4 uses the operator-norm control at exactly that point: without it the classes  $[\text{Ad } U_n(g)]_\omega$  need not be multiplicative.

Amplification by an identity block separates the two normalizations. Padding a model with an identity block preserves every operator-norm quantity — the multiplicative defects and the pairwise separations are unchanged — while the normalized Hilbert–Schmidt distances decay with the padding ratio: below any prescribed bound, every approximate unitary representation can be replaced by one all of whose pairwise Hilbert–Schmidt distances are smaller than the bound. 2-norm triviality of an element therefore does not by itself prove a group non-MF: an obstruction must produce an invariant corner with its own normalized trace, so that the normalization is unchanged by amplification. The relative rank of the corner is unconstrained: padding leaves the averaging corner  $q_n M_{d_n}(\mathbb{C}) q_n$  untouched while  $d_n$  grows, so  $r_n/d_n$  may tend to zero while the argument uses the corner’s own normalized trace.

The spectral projection of Theorem S5.6 is normalized on its own range in the same way, and the proof of Theorem S10.1 normalizes by the rank  $k_n$ .

Finite dimensionality of the coordinates cannot be replaced by stable finiteness of the coordinate algebras together with faithful traces. For faithfully traced unital  $C^*$ -coordinates  $A$ , the left and right actions on  $L^2(A, \tau)$  generate a von Neumann algebra that need not be finite: for an infinite-dimensional  $\text{II}_1$  factor in standard form it is all of  $B(L^2(A, \tau))$ . Constant coordinates  $A_n = C_r^*(E)$  give a counterexample to any such generalization: the left regular representation embeds  $E$  exactly and injectively into the norm quotient

$$\prod_n A_n / \bigoplus_n A_n,$$

the element  $c$  centralizes the Kazhdan base there, and conjugation by  $t$  produces  $tct^{-1}$ , whose commutator with the base is the element  $u$  with  $u^2 = w \neq 1$  of Theorem A. Precisely: Theorem 3.4 becomes false when its matrix coordinates are replaced by stably finite unital  $C^*$ -algebras carrying faithful tracial states and its normalized Hilbert–Schmidt norm by the trace 2-norm, with all its remaining hypotheses unchanged.

Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]; that question is independent of Theorem 3.4. The adjoint spectral estimates used here need operator-norm multiplicative control, which permutation and Hilbert–Schmidt models do not supply; soficity of  $E$  comes instead from its normal form (Theorem D).

### Questions.

- (1) What is  $\text{Res}_{\text{MF}}(E)$ ? It properly contains  $\{1, w\}$  by Theorem B, and Corollary S10.5, applied to  $E/\langle w \rangle$  and its collapse defect  $D$ , computes it from  $\text{Res}_{\text{MF}}((E/\langle w \rangle)/D)$ . Neither of these two specific collapse applications gives further information about that second quotient:  $w$  is already trivial in  $E/\langle w \rangle$ , and the commutators generating  $D$  are already trivial in the quotient by  $D$ . This does not rule out some other finite-normal, normal-Kazhdan or orbit-collapse datum in it.
- (2) Is there a torsion-free finitely presented non-MF group? Every finite subgroup of a torsion-free group is trivial, so Theorem S5.3 is empty there, while Theorem S5.6 applies to torsion-free groups. What remains is group-theoretic: a torsion-free finitely presented group whose subgroup  $N_{\text{conj}}$  contains a *nontrivial* normal property-(T) subgroup. The trivial subgroup is normal and has property (T) vacuously and gives no obstruction. Remark 5.2 gives a small-cancellation route over the torsion-free finitely presented Kazhdan group of [FFF], which removed the analogous torsion obstacle for nonsoficity. The Clifford construction of Theorem 5.1 cannot produce such a group: it imposes  $c^2 = 1$ , so  $d^2 = 1$ , and a torsion-free quotient  $q$  then has  $q(d) = 1$  and  $q(N_{\text{conj}}) = 1$ .

- (3) Does MF imply hyperlinearity? A faithful embedding into a matrix quotient need not make any coordinate trace faithful, so the defining MF representation does not immediately give a hyperlinear model. This question concerns the CDE convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.
- (4) For which groups do the criteria of Theorems S5.3, S5.6 and S10.1 compute the residual exactly?
- (5) Does the obstruction survive when the matrix blocks are replaced by other building blocks? The proof of Theorem 3.4 uses finite dimensionality of  $M_{d_n}(\mathbb{C})$  twice: for the conjugation action on  $L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$ , and for finiteness of the norm ultraproduct. Faithful traces alone do not replace it. For which separable  $C^*$ -algebras  $A_n$  does every homomorphism  $H \rightarrow \mathcal{U}(\prod_n A_n / \bigoplus_n A_n)$  still map  $w$  to the identity? The lifting of Lemma 2.1 is *not* where the matrix structure enters, and it is not a semiprojectivity question: for unital  $A_n$  and  $u$  unitary in the quotient, any lift  $(x_n)$  has  $\|x_n^* x_n - 1\| \rightarrow 0$  and  $\|x_n x_n^* - 1\| \rightarrow 0$ , so  $x_n$  is invertible for all large  $n$  and the polar correction  $u_n = x_n(x_n^* x_n)^{-1/2}$  is unitary with  $\|u_n - x_n\| \rightarrow 0$ . Neither real rank zero nor semiprojectivity in the sense of Loring [Lo98] is used. The two genuinely matrix-specific ingredients are the ones listed above.

#### S14. THE MAXIMAL GROUP $C^*$ -ALGEBRA

For a countable group  $H$ , the maximal group  $C^*$ -algebra  $C_{\max}^*(H)$  is the completion of the group ring  $\mathbb{C}[H]$  in the norm

$$\|x\|_{\max} = \sup_{\pi} \|\pi(x)\|,$$

the supremum taken over unitary representations of  $H$ ; the supremum is finite because  $\|\pi(\sum_h \lambda_h h)\| \leq \sum_h |\lambda_h|$ . Write  $u_h$  for the canonical unitary of  $h \in H$ .

**Proposition S14.1** (the universal property). *The canonical map  $h \mapsto u_h$  is injective, and for every unital  $C^*$ -algebra  $B$  lying in a fixed universe that contains  $H$ , and every homomorphism  $\rho: H \rightarrow \mathcal{U}(B)$ , there is a unique unital  $*$ -homomorphism  $C_{\max}^*(H) \rightarrow B$  with  $u_h \mapsto \rho(h)$  for every  $h \in H$ .*

Injectivity comes from the left regular representation, uniqueness from density of  $\mathbb{C}[H]$ . Restricting  $B$  keeps “all representations” a set, and every target below is a unitary group of a separable  $C^*$ -algebra.

**Proposition S14.2** (a strictly dominated projection gives a proper isometry). *Let  $A$  be a unital  $C^*$ -algebra containing a projection  $p$  and a unitary  $u$  such that the conjugate  $upu^*$  satisfies  $p(upu^*) = (upu^*)p = p$ , equivalently  $p \leq upu^*$ , and  $upu^* \neq p$ . Then  $A$  contains an isometry that is not a unitary; consequently  $A$  is not stably finite and has no faithful tracial state.*

*Proof.* Write  $q = upu^*$ , so  $q$  is a projection with  $pq = qp = p$ , and put  $r = pu^*$ . Then

$$r^*r = uppu^* = upu^* = q, \quad rr^* = pu^*up = p.$$

Put  $s = r + (1 - q)$ . Since  $p(1 - q) = 0$  and  $(1 - q)p = 0$ ,

$$s^*s = r^*r + (1 - q) = q + (1 - q) = 1,$$

and since  $pu^*(1 - q) = pu^* - pu^*upu^* = pu^* - pu^* = 0$  and likewise  $(1 - q)up = 0$ ,

$$ss^* = rr^* + (1 - q) = p + (1 - q) = 1 - (q - p).$$

Thus  $s$  is an isometry, and  $ss^* = 1$  would force  $q = p$ , which is excluded; so  $s$  is not a unitary. A unital  $C^*$ -algebra containing a nonunitary isometry is not finite, hence not stably finite. If  $\tau$  were a faithful tracial state then  $\tau(1 - ss^*) = \tau(1) - \tau(s^*s) = 0$  while  $1 - ss^* = q - p$  is a nonzero positive element, contradicting faithfulness.  $\square$

*Remark S14.3* (the maximal algebra under proper one-sided conjugation). Let  $G$  contain a property-(T) subgroup  $\Gamma$  and an element  $t$  with  $t\Gamma t^{-1} \subsetneq \Gamma$ ; the base of  $E$  is such a pair. Let  $P \in C_{\max}^*(G)$  be the spectral projection at 1 of the Hermitian Kazhdan average  $\frac{1}{2|S|} \sum_{s \in S} (u_s + u_s^*)$ , which exists by the spectral gap of Section 3. Then  $P \leq u_t P u_t^*$ , and the inequality is strict: in the quasi-regular representation on  $\ell^2(G/t\Gamma t^{-1})$  the point mass at the base coset is fixed by  $t\Gamma t^{-1}$  and moved by every element of  $\Gamma$  outside it, so the two projections differ there. By Proposition S14.2,  $C_{\max}^*(G)$  is then not stably finite, has no faithful trace, and in particular is neither residually finite-dimensional nor MF. For the reduced algebra of  $E$ , by contrast, the canonical trace is faithful and the algebra is stably finite (Theorem C), so its failure of MF is not a failure of finiteness.

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